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Body data-driven garment pattern construction in digital fashion innovations: a review

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Abstract

Body data are indispensable for garment pattern construction, not only influencing garment quality but also shaping the competitiveness of apparel enterprises and driving the digital transformation of the fashion industry. Traditional pattern construction practices primarily rely on a limited number of one-dimensional (1D) measurements, with less consideration given to higher-dimensional body data, such as three-dimensional (3D) shapes and four-dimensional (4D) motion information. This underutilization of advanced anthropometric data reflects the technological limitations of past eras. With the rapid advancement of digital technologies, cutting-edge anthropometry and digital tools have significantly enhanced the integration of body data into pattern construction. To comprehensively understand the utilization of body data in digital pattern construction, as well as to clarify the current technological landscape and future directions, this review provides a thorough analysis of the evolution of body data utilization and facilitators in pattern construction, pattern construction methods based on multidimensional body data (1D–4D), as well as the distinctions and applications of these approaches. Furthermore, this review offers a comprehensive examination of the prospects and challenges in pattern construction, considering key aspects such as dynamic pattern construction, AI-driven generation, CAD system integration, and emerging business opportunities. It underscores the scientific and innovative potential of high-dimensional 4D body data and AI-powered tools in advancing modern garment pattern construction. By highlighting these developments, this review aims to provide researchers with valuable insights, enabling them to anticipate and explore uncharted possibilities in the field.

Keywords

Body data, pattern construction, body measurement, 4D data, dynamic, CAD, artificial intelligence, AI generation

Leveraging body data for pattern construction has consistently been a core technical challenge in the apparel industry. Due to the inadequate and improper utilization of body data, garment fit issues and multiple rounds of pattern iteration are inevitable,[1] placing significant pressure on fashion companies both on cost and time. To enhance the efficiency and precision of garment pattern construction, the fashion industry has embraced advancements in digital anthropometry and computer technology, integrating them innovatively.

Anthropometric technologies provide a fundamental basis for garment pattern construction, and the rising interest in digital anthropometry aligns closely with digital fashion innovations. Typically, body data can be categorized into one-dimensional (1D), two-dimensional (2D), and three-dimensional three-dimensional (3D),[2] with respective data carriers being body measurements, body cross-section curves, and point clouds. These forms of body data are static, as they represent measurements captured from a standard, fixed posture.[1] To gain a more comprehensive understanding of the human body, further research is required from a dynamic perspective. Conversely, four-dimensional (4D) body data refers to dynamic human body data that varies over time within 3D space. This type of data not only include the shape

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and dimensions of the body but also account for the changes in body morphology during movement and the dynamic characteristics related to time. They are represented as a sequence of 3D scans taken over time.[3] Notably, manual or 3D static recordings of body measurements in various postures are often considered dynamic measurements, focusing on analyzing movement-related changes in body measurements.[1] Although not strictly classified as 4D data, these measurements incorporate body movement information and are widely used in functional wear design, an area of significant research interest before the advent of 4D technologies. Figure 1 presents four different types of human body data.

The data utilized in pattern design gradually change from static body data to dynamic body data. Traditional pattern construction methods rely solely on body measurements, with insufficient consideration of the body's varying morphological features.[8] To implement a human-centered design concept, designers must accurately interpret and apply body data to garment pattern design to ensure optimal fit and functionality. When higher-dimensional body data, such as 3D or even 4D data, are employed in pattern construction, the accuracy in approximating the actual contours of the human body is improved significantly.

4D body data have transcended the limitations of traditional pattern construction rules based on static measurements. While conventional methods primarily aim for a good fit in static postures, 4D pattern design emphasizes continuous fit throughout dynamic motion. By analyzing changes in skin dimensions, deformation rates, and soft tissue vibrations during various movements, 4D-based approaches enable the development of garment patterns that better accommodate dynamic conformity. This provides critical support for the design and optimization of functional and performance apparel, as well as ergonomic product design and evaluation. Looking ahead, the integration of 4D data with advanced virtual simulation systems could allow designers to directly observe garment performance in motion and make real-time pattern adjustments. Moreover, 4D scanning technologies offer promising prospects for high-level customization, particularly in elite sportswear, functional clothing, and rehabilitative medical garments.

The tools that facilitated the pattern construction also changed with social innovation. To reduce manual work involvement, body data in an appropriately formatted input file are imported into computer-aided-design (CAD) software. CAD systems accelerate product development and reduce the time from design

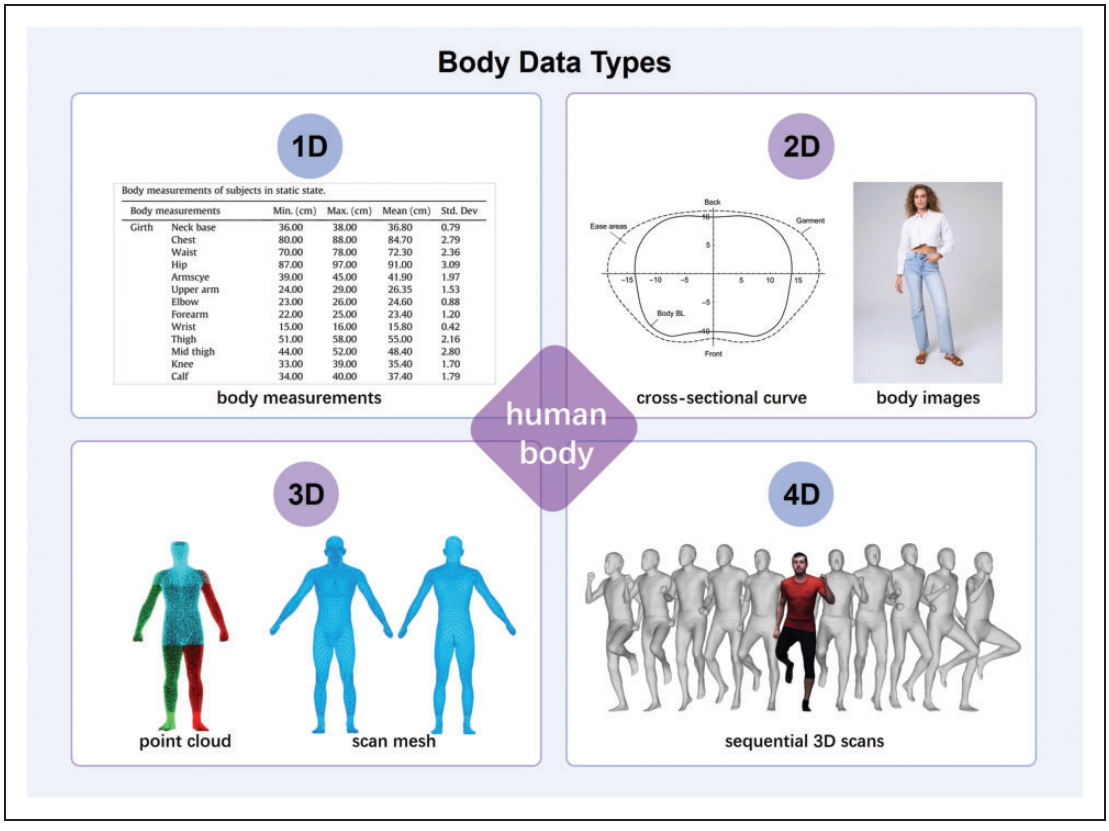


Figure 1. Body data types.[4–7]

to manufacturing,[9] enabling remote design or codesign between customers, pattern makers, and manufacturers. In order to move toward a high level of intelligence, apparel CAD needs to introduce artificial intelligence technologies.[10] Recently, AI has empowered fashion design in various aspects. It can analyze and learn from vast amounts of body and garment data, demonstrating exceptional performance in body and garment size prediction,[11] pattern generation, virtual garment generation, and virtual fitting. These facilitators help bridge the gap between extracted body data and pattern construction.

AI tools offer distinct advantages in pattern construction compared to traditional methods. Manual pattern construction is highly dependent on human labor, whereas conventional CAD systems merely provide drawing interfaces and still rely on the expertise of experienced pattern makers. In contrast, AI tools combine intelligent decision-making with learning capabilities, enabling them to automatically generate pattern designs by learning from extensive datasets that include pattern, body data, and fabric properties. Furthermore, the application prospects of AI in this domain are broad and promising. Through deep integration with 3D virtual fitting technologies, AI facilitates rapid validation of pattern effectiveness, thereby accelerating the product development process. In addition, AI-generated pattern data can directly support intelligent cutting and flexible production systems, promoting the transition toward smart manufacturing in the apparel industry.

At present, several reviews related to garment pattern construction have been conducted within the context of digital technologies. Yang et al.[12] reviewed pattern-making studies from the perspective of 2D drafting and 3D flattening methods. Sayem et al.[13] focused on 3D CAD systems for the clothing industry. Liu et al.[14] reviewed core technologies in 3D garment design. These review articles focused on either garment CAD system application, 3D garment design, or 2D/3D CAD methods. Yet, none have clarified how body data are integrated into the subsequent stages of the pattern construction workflow. Moreover, earlier reviews exhibit certain temporal limitations, failing to encompass cutting-edge developments in 4D body data and AI-driven pattern construction applications. These two aspects represent both key challenges and research frontiers for the future, offering significant insights for academic research in the field of apparel.

By contrast, this review explores the evolution and advancement of pattern construction techniques through two primary lines, encompassing cutting-edge research on both 4D body data and the application of AI tools. The first line examines the development and application of human body data, progressing from 1D body data to exploratory applications of 4D body data in pattern construction, reflecting the industry's deepening understanding of the human body and the shift from static to dynamic studies. The second line addresses the evolution of crucial tools supporting garment pattern construction, transitioning from manual tools to CAD and now to AI, which offers new creative potential in pattern design. Furthermore, this review intersects these lines, summarizing the methods, strengths, and limitations of different dimensional human body data in garment pattern construction. Finally, it presents the challenges and future outlook to provide insights for academics and professionals engaged in garment pattern construction. Figure 2 illustrates the overall structure of the review. Figure 3 illustrates pattern construction methods utilizing different dimensional body data.

Evolution of body data utilization and facilitators in pattern construction

Evolution of body data utilization in pattern construction

1D body data. 1D body data are linear measurements; they can be obtained by manual measurements, 3D scans, and body images. First, the manual process of taking anthropometric data is time-consuming and lacks reproducibility.[15] Measurement tools such as measuring tapes, anthropometers, and calipers are predominantly applied in manual measurements. Confronted with the complex morphology of the human body, skilled technicians take measurements by referencing specific anatomical landmarks on the body, ensuring accuracy and consistency. Second, measurements can also be extracted from 3D body scans, such as height, chest circumference, and waist circumference. These data can be further consolidated into size charts and sizing systems for standardization. Third, image-based methods capture images of the human body from orthogonal perspectives, typically from the front and side views.[14] Using mathematical models, basic dimensions such as heights, length, and circumferences can be estimated from photographs.[16] However, the number of collected body dimensions remains relatively limited, and the approximated circumferences may exhibit reduced accuracy.[16]

1D body data represent human body characteristics in a highly simplified form, with an emphasis on size-related parameters. The most commonly used pattern-making approach involves first calculating the regression relationships among key body dimensions, followed by the development of flat pattern-making formulae and grading rules. Based on these, flat patterns that correspond to standard body sizes are constructed.

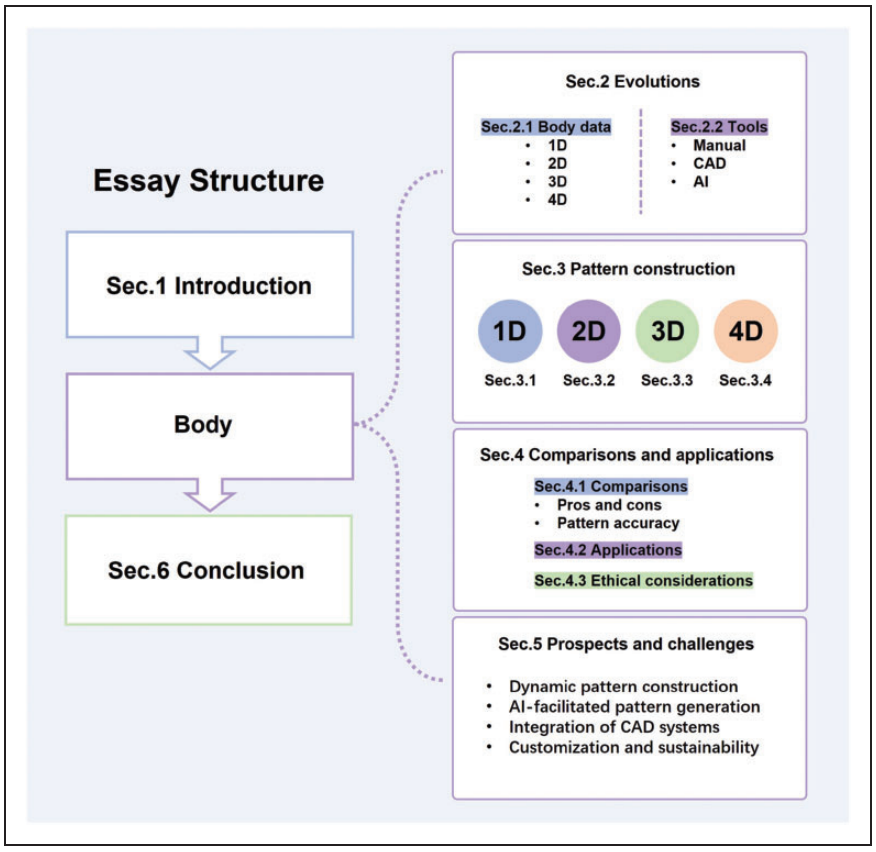


Figure 2. Overall structure of the review.

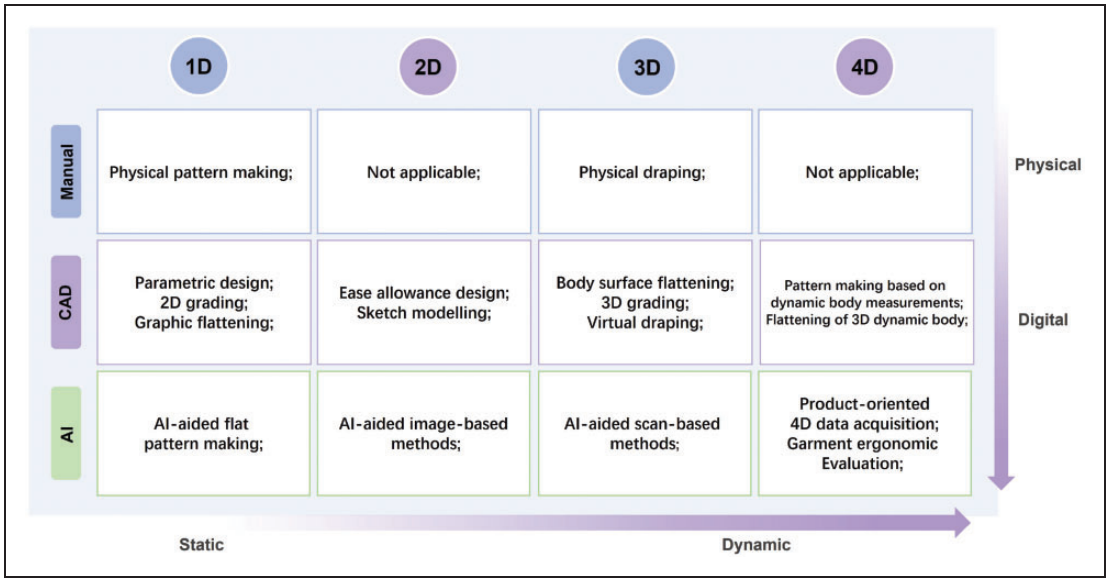


Figure 3. Body data utilization in pattern construction.

2D body data. Admittedly, 2D body data are intrinsically linked to their 3D foundation and are typically regarded as body cross-sectional data or body outline silhouettes. This review also includes 2D body images in the discussion, as they represent a crucial 2D medium of body information in pattern-related studies.

The transverse planes on the bust, waist, and hip, as well as the sagittal plane, created the most important

cross-section curves, which contain essential information for basic pattern generation. Body cross-sectional data were explored extensively in garment pattern ease allowance research. Traditional manual measurement struggles to capture cross-sectional data. In contrast, with the development of 3D body scanning, acquiring 2D body cross-sectional information has become more accessible. Lu and Wang[17] utilized a convex hull polygonal approximation method to calculate the body cross-section circumferences from point cloud data. The circumference points were obtained by slicing the point cloud with a perpendicular plane.[18] Cubic-spline curves[5] and elliptic Fourier[19] are also utilized in body cross-section or silhouette curve fitting. Noteworthily, body outline silhouettes provide an intuitive reference for designers when creating garment sketches during the creative design process. In addition, with the assistance of AI, 2D body images, whether with or without clothing, play a key role in body measurement extraction, virtual garment generation, and pattern construction.

2D body data provide intuitive information on body contours, assisting pattern makers and designers in accurately understanding the shape and curvature of different body regions. In pattern design, 2D body data serve as an intermediary medium, either facilitating the estimation of 1D measurements or supporting the generation of 3D garment models, which can then be further utilized for pattern development.

3D body data. 3D scanning is fast, reliable, and highly reproducible, with a lower variance, making it easier to understand different morphologies and providing better comfort and fit for garment products. It enables the rapid extraction of extensive human body data within seconds. According to the technical principles, 3D scanners are categorized into passive stereo, structured light, and time-of-flight imaging.[18] These systems can also be classified into stationary, handheld, and mobile scanners, with the latter two addressing the drawbacks of high costs and limited portability.

The raw data generated by the scanning system can be represented in various formats, including 3D point clouds, depth maps, and 2D images. After processing through specialized software, the output data may take forms such as point clouds, meshes, or body measurements.[15] Human body modeling enables the acquisition of more detailed information compared with unstructured point cloud data sets.[20] Reverse engineering software, such as Geomagic, Rapidform, and Imageware, can read, visualize, reconstruct, and measure these point clouds. During this process, point data are converted into a mesh, undergoing operations such as smoothing, decimation, and hole filling.[15] Common approaches for reconstructing a continuous body model from point clouds include implicit methods, parametric methods, and triangulation-based methods.[14] In addition, scanning systems are integrated with measurement extraction tools, enabling automatic data retrieval from raw scanned files, and this topic has become a recent research focus.[21] However, 3D scanning also presents limitations, particularly in capturing nonvisible regions such as the underarm or inner thigh, where occlusions may hinder accurate data acquisition.

When the scanned body is minimally clothed, such as in underwear or tight-fitting garments, the 3D body contours are fully revealed. This comprehensive 3D body scan data significantly enhance the precision of tight-fitting garment pattern designs, such as underwear, leggings, and compression suits, since tight-fitting garments can be represented as displacements from the body skin. Cheng and Kuzmichev[22] analyzed scanned male lower body morphology to improve men's underwear design. Hong et al.[23] utilized the atypical morphology of scoliosis patients to design customized garments. In this context, 3D body data greatly facilitate the personalized and customized design of garments.

When processing body scans with loose-fitting garments, garment reconstruction techniques, body-garment segmentation methods, 3D garment model cutting line design, and surface flattening are essential for obtaining accurate garment patterns. In the early stages, reconstruction techniques primarily relied on geometry-based processing, for example, Chen et al.[24] employed the Bowyer–Watson algorithm to perform Delaunay triangulation for the surface reconstruction of garment point cloud data, the corresponding flattened pattern was then derived using a finite-element (FE) approach. In recent years, significant breakthroughs have been made in deep-learning (DL)-based 3D reconstruction techniques, fundamental techniques are neural radiance fields (NeRFs), latent diffusion models (LDMs), and 3D Gaussian splatting.[25] Due to the presence of noise, garment shape variability in scan results, and the difficulty of separating clothing from the human body, the postprocessing of models becomes highly complex. As a result, 3D reconstruction based on real-scanned garments has limited practical applications in areas such as pattern generation and virtual garment fitting.

In a nutshell, 3D body data capture the full spatial geometry of the human body, enhancing both the efficiency and visualization of garment pattern design. As the digitization of pattern construction advances, various forms of 3D body data are playing an increasingly critical role in 3D virtual pattern design and virtual fitting-based fit validation. Moreover, 3D body data of target consumer groups have become a valuable

digital asset for apparel companies across different market segments.

4D body data. As clothing products must accommodate changes in size and shape during wear due to human movement, the application of dynamic anthropometric technology to acquire 4D human body data has become a critical topic in functional apparel research. The real meaning of 4D is sequential 3D. 4D data provide in-depth insights into movement patterns and morphological changes that cannot be observed in a static 3D state, as well as the interactions between clothing and the human body, thereby enhancing researchers' comprehensive understanding of body deformation during motion. Current research on 4D human body data primarily focuses on areas where soft tissue deformation or vibration is most pronounced during movement, such as the legs, breasts, and feet. These findings are widely applied in the human-centered design of functional apparel products, including sportswear, sports bras, protective clothing, and athletic footwear.

Studies in this field have successively employed (a) multi-posture 3D scanning, (b) motion capture technology, (c) 4D scanning technology, and (d) AI-generated 4D data, as illustrated in Figure 4.

(a) Multi-posture 3D scanning

Multi-posture 3D scanning serves as a prelude to 4D research, capturing several body postures to analyze single frames of specific dynamic postures. This technique aims to obtain 3D data reflecting human motion characteristics, which can be used as a reference for designing sportswear and functional clothing that best adapts to skin deformation.

One primary research direction is the measurement of body surface dimension variation rates. The standard standing scanning posture is typically used as a reference, the analysis aims to investigate the dimensional discrepancies between static and dynamic postures and to summarize the patterns of skin deformation.[27,34–38] For some dynamic postures that are difficult to scan or reducing the number of scanning postures, some studies employ parametric 3D human models, such as SMPL,[39] to develop postured dynamic body models.

Another research direction focuses on mapping surface tension lines of the human body. Based on the dimensional differences observed between static and dynamic postures, researchers map dynamic tension lines[40] and lines of non-extension[41,42] on the skin surface in specific postures. These mappings serve as references for designing garment segmentation lines.

Table 1 summarizes the representative literature on multi-posture 3D scanning in the field of clothing in recent years. Although multi-posture 3D scanning can partially reflect human motion changes without requiring expensive 4D equipment, it still presents notable limitations compared to real 4D data. First, 3D body scanners are typically designed for static standing postures and cannot automatically capture dynamic dimensions, requiring manual extraction by researchers. Consequently, the accuracy of dynamic dimension measurements is heavily dependent on the precise localization of body markers. In addition, individual differences in body composition and inconsistencies in executing target postures affect the reproducibility and comparability of measurement results. Furthermore, maintaining a single posture for an extended period involves muscle contraction and relaxation, deviating from the natural continuity of human motion.

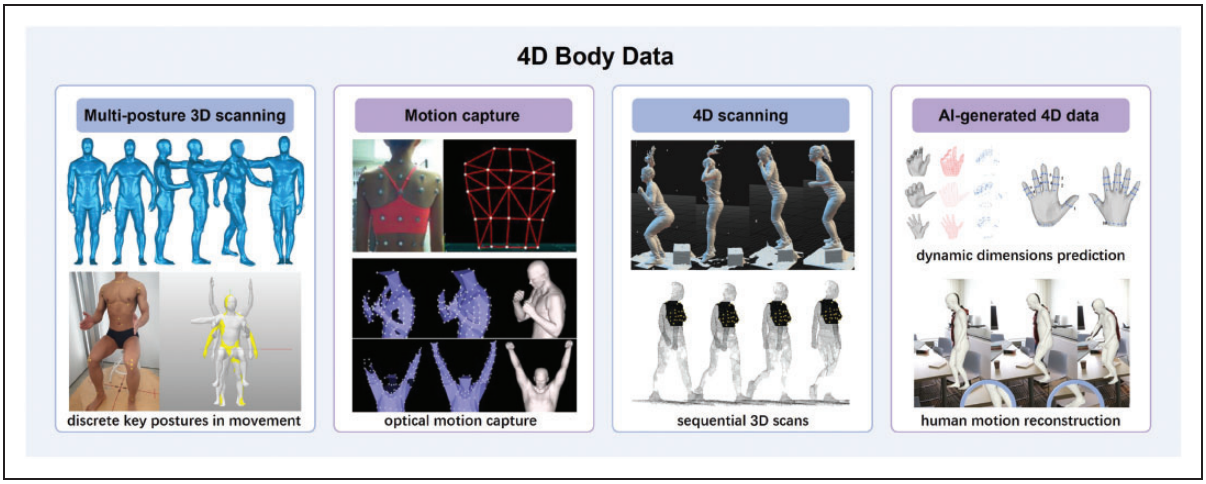


Figure 4. 4D body data acquisition.[26–33]

Table 1. Typical multi-posture 3D scanning research in recent years

Study	Year	Postures
Zare et al.[38]	2025	Hands in static and five defined dynamic postures.
Kang and Kim[27]	2023	Standing posture, five defined dynamic postures.
Loercher and Morlock[35]	2018	Standing posture, nine defined dynamic postures.
Petrak and Naglic[36]	2017	Standing posture, two defined dynamic postures.
Choi and Hong[41]	2015	Standing posture, knee joint flexion of 30°, 45°, and 60°.
Shin and Chun[37]	2013	Standing posture, four baseball batting postures.
Lee et al.[42]	2013	Standing posture, 90° knee flexion, squatting.
Seo et al.[40]	2013	Standing posture, knee joint flexion of 30°, 45°, and 60°.
Choi and Ashdown[34]	2011	Standing posture, 120° knee bend posture, one pace stepping posture, sitting posture.

(b) Motion capture technology

Motion capture technology similarly provides information on temporal changes in human body dimensions. There are various motion capture technologies based on different principles, including optical, inertial, and electromagnetic systems. Among these, optical motion capture offers the highest measurement accuracy and is widely utilized in the field of dynamic anthropometry. Optical motion capture requires the attachment of markers to the human body, which then serve as a sparse representation of the body's shape. For example, Sohn and Bye[28] measured body surface measurement change in motion using an optical motion capture system.

However, the positional changes of markers during movement may introduce measurement errors. More importantly, the 4D human body data captured by motion capture technology only represent the motion of specific markers or skeletal structures, rather than the dynamic changes of the entire body surface. To address this limitation, researchers have integrated motion capture data with 3D scanning data to generate digital human models in motion, thereby obtaining comprehensive surface movement information. One 4D measurement method is based on dense static shape scanning and sparse dynamic motion capturing,[43] rigging 3D body scans by using an internal skeleton to generate certain body movements. For example, Park and Hodgins[29] utilized optical motion capture data to rig scanned body models, deforming a subject-specific model to align with the trajectories of the markers, thereby analyzing the deformation of body parts during specific movements. Pirch et al.[44] used 3D scanning to create 4D motion data for clothing simulation.

(c) 4D scanning technology

Since 3D scanners only allow precise scanning of static positions and cannot capture dynamic body movement,[15] body scanning technology has integrated real-time motion capture capability, commonly called 3D scanning in motion or 4D scanning. The technical principles of 4D scanning systems are structured light or stereo vision. 4D scanning systems integrate the advantages of 3D scanning and motion capture technologies. Representative commercial 4D scanning systems include 3dMD,[45] MOVE4D,[7] and Hohenstein 4D scanner, as shown in Figure 5.

This development has brought increasing attention to 4D body data. The 4D scanning technology is capable of recording point cloud data for each frame of human motion, yielding substantially larger and more detailed information, which can be stored in various file formats. Considering that all wearable products are subjected to body movement or motion, this progression is driving garment pattern design toward greater comfort and enhanced functionality. For example, the Move4D[7] scanning system captures full-body movement over time and generates homologous meshes. The corresponding processing software relies on DL and data-driven human body models, incorporating elements such as shape, posture, and soft tissue deformation.

Precise and automated tracking of anthropometric landmarks marked on a subject's body surface can enhance the analysis of dynamic anthropometric data obtained through a 4D scanning system.[46]

Zhang et al.[47] employed the 3dMD foot scanning system to analyze dynamic foot shape changes during the foot roll-over process, Tajdari et al.[48] developed the open-access 4D-Feet dataset and introduced a framework to synchronize spatiotemporally asynchronous scans captured from multiple cameras and track of point correspondences across all frames, facilitating the extraction of dynamic features for each vertex. This research provides valuable data for optimizing footwear design. Liu et al.[31] integrated 4D scanning with motion capture to improve interframe correspondence, enabling a more precise investigation of the dynamic properties and deformation patterns of the breast surface during movement. Kuehn and Kyosev[49] captured a motion of a men moving both

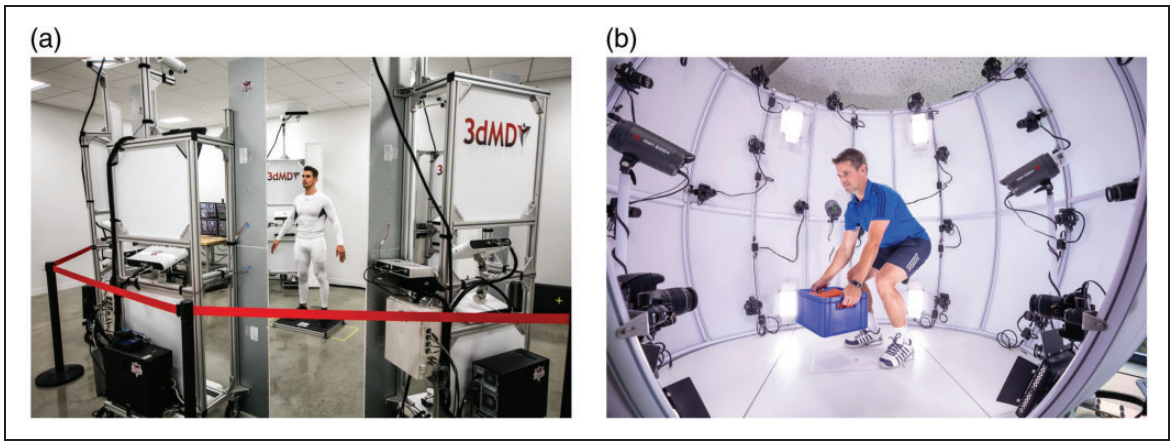


Figure 5. Commercial 4D scan systems: (a) 3dMD in Under Armour (https://3dmd.com) and (b) Hohenstein 4D scanner (https://www.hohenstein.com).

Table 2. Typical 4D body data studies and potential application areas

Study	Year	4D equipment	Region-specific analysis	Potential application areas
Tajdari et al.[48]	2024	TU Delft 4D foot scanner	Feet	Sports footwear or insoles
Liu et al.[31]	2024	3dMD	Female breast	Sports bra
Schmidt et al.[50]	2024	MOVE4D	Full-body figure	Dynamic virtual fitting
Schmidt and Kyosev[51]	2023	MOVE4D	Female breast	Sports bra
Zhang et al.[47]	2023	3dMD foot scanner	Feet	Sports footwear or insoles
Kyosev et al.[52]	2022	MOVE4D	Leg	Compression leggings
Pei et al.[53]	2021	3dMD	Female breast	Sports bra
Kuehn and Kyosev[49]	2021	MOVE4D	Full-body figure	Men's formal shirt and trousers

hands from down to up by MOVE4D, then computed point cloud distances between the frames of the men's shirt, their analysis revealed that the point cloud distance accurately reflects the actual slippage and sticking behaviors of the garment.

Furthermore, the integration of 4D data with SMPL human models and FE models has opened new avenues for research in human body modeling and human–garment interaction. For example, Schmidt et al.[50] proposed an automated approach for fitting skeletal structures to 4D scan data by leveraging SMPL models and the OSSO Python module, achieving efficient integration of skeletal frameworks with surface geometry. This method demonstrates strong potential for applications in virtual garment fitting. Schmidt and Kyosev[51] utilized 4D scan data to construct a FE model simulating the interaction between a bra and the female breast, highlighting the potential of 4D data for dynamic simulation of soft material interactions in apparel and textile applications.

Table 2 summarizes representative studies on 4D data in recent years and their potential application areas.

Despite their ability to capture high-resolution, high-frequency data of moving bodies, 4D scanning systems present substantial challenges in data processing. Postprocessing techniques tailored specifically for 4D data remain in their infancy. Key technical bottlenecks include the lack of automated analytical methods, the absence of accurate and temporally consistent dense correspondences between frames,[31] and the non-standardized file formats for storing complete 4D sequences, which lead to poor interoperability across different systems. To mitigate these issues, current research strategies often involve analyzing key frames that represent critical postures, thereby reducing computational demands. In addition, trackability is improved through the application of body surface markers or the use of homogeneous 3D mesh processing techniques, the latter being viable primarily when subjects wear tight-fitting garments or underwear.[54]

(d) AI-generated 4D data

AI-generated 4D data leverage data-driven approaches by utilizing dynamic human body datasets established through motion capture and 4D scanning technologies, such as Dyna,[55] Dynamic FAUST,[56] and AMASS.[57] These high-quality motion capture and 4D

scan datasets enable the learning of motion priors, providing a fundamental basis for DL-based 4D data generation. AI-generated 4D data have been explored in two key areas: predicting dynamic body dimensions and deformations, and learning as well as reconstructing human motion.

First, in AI-based prediction of dynamic dimensions and human body deformation, Dai et al.[32] proposed Measure4DHand, the first DL-based dynamic measurement method. This model automatically extracts dynamic hand dimensions from point cloud sequences scanned by Kinect, facilitating the analysis of size variations caused by skin deformation under different postures. This approach provides valuable insights into the functional design of hand-related products.

Moreover, in AI-based learning and reconstruction of human motion, researchers have developed various techniques to enhance motion analysis. For instance, ClothCap[58] is a data-driven approach that segments dynamic clothing and the human body from 4D scan data, allowing their retargeting onto new human models. Similarly, LEMO[33] learns motion priors from the AMASS motion capture dataset to achieve human motion recovery from RGB(D) sequences. Furthermore, 4D Gaussian splatting[59] enables dynamic 3D scene reconstruction by deforming a set of canonical 3D Gaussians over time, thereby producing motion-aware 3D models. This technique holds significant potential for advancing research on dynamic human body modeling.

To summarize, digital body data captured in motion reflect changes throughout the entire movement, allowing for the extraction of anthropometric data at various stages. The use of 4D body data can translate the body's deformable qualities into patterns, thereby increasing freedom of movement, making it particularly valuable in the design of sportswear. 4D-based pattern design places greater emphasis on the body's movement patterns within real-world dressing scenarios, aligning pattern design more closely with the garment's intended end use. Furthermore, the integration of 4D data into ergonomic analysis and dynamic virtual fitting has become a key approach for optimizing dynamic patterns.

To provide a clear and concise summary of this section, Table 3 outlines the concepts and applications of the four types of body data discussed.

Evolution of facilitators in pattern construction

Tools for manual pattern construction. Various tools are involved in physical pattern construction. For manual flat pattern making, pattern makers use measuring tapes to take body measurements, along with drafting tools such as pattern paper, straight rulers, chalk, scissors, and tracing wheels.[60] In terms of draping, mannequins representing target customers are used, along with tailor pins. Fabric is directly applied to the mannequin, and pattern makers manipulate the cloth using techniques such as cutting, folding, and pinning. Draping offers the advantage of capturing the interaction between the body and the garment, and inspires practitioners' creativity.

Traditional pattern construction processes have been conducted in the industry by expert manual know-how[61] and require significant time and specialized skills,[60] and this process continues to be embraced by couture brands, highlighting the enduring significance of time-tested techniques. However, managing physical pattern files is challenging, therefore, garment pattern information cannot be systematically integrated to form the company's assets.

CAD systems for digital transformation. With the rapid evolution of computer technology, pattern construction has gradually transitioned from manual drafting to CAD. 2D CAD software facilitates the drafting of geometric patterns based on anthropometric measurements of the target's size and shape. 3D CAD software enables two distinct methods in garment design: the "2D to 3D" approach, which refers to draping digital 2D pattern pieces onto a 3D mannequin, and the

Table 3. Body data types and applications

Types	Concept	Applications
1D	Linear measurements, such as lengths, circumferences, and widths.	Determine the basic dimensions of garment, for instance, bust measurement defines the size of shirt.
2D	Contours that capture the shape of the body in two planes, such as body outline contours and cross-sectional profiles.	Create flat patterns that consider body curvature and shape.
3D	Encompass volumetric measurements and spatial coordinates.	Create more accurate garment patterns that perfectly contour to the body's unique shape.
4D	3D body data combining time as the fourth dimension, capturing how the body moves and changes shape over time.	Essential for designing high-performance and functional garment pattern that need to accommodate dynamic body movements.

"3D to 2D" approach,[13] which implies the development of garment on a 3D body and further flattening the shape into 2D pattern pieces. The cross-use of 2D/3D CAD systems has optimized traditional garment pattern design processes.

In contemporary academic research on garment construction, researchers have developed specific CAD systems, such as JBlockCreator[21] and DisDraw.[62] In addition, commercially available 2D CAD systems, such as AutoCAD,[16] Gerber CAD,[63–65] and 3D CAD systems, including CLO 3D, Style 3D, and Vstitcher, have been utilized for pattern construction. In both academic and commercial systems, these platforms facilitate the input of 2D body measurements or the import of 3D body models. Certain CAD systems[16] seamlessly connect the pattern design workflow with advanced anthropometric technologies, enhancing automated garment pattern construction. In addition, advanced 3D systems offer a high degree of interactivity, where any modification to the 2D patterns automatically updates the corresponding 3D design.[66]

AI as future power. AI tools have revolutionized garment pattern design, driving sustainability, personalization, and innovation in the fashion industry. In general, AI enhances CAD by enabling real-time decision-making through quantitative reasoning, simulations, and deep modeling.[67] Evolving from analytical to generative AI,[68] these technologies enhance both predictive accuracy and automated pattern creation. This section reviews AI applications in two areas: (a) AI-driven pattern parameter prediction, which uses AI to estimate garment dimensions, fabric behavior, fit, and more; (b) AI-generated pattern design (AIG-Pattern), leveraging DL and generative models for automated pattern creation and optimization. The following sections will explore the role of machine learning (ML) in predictive modeling and the contribution of DL to generative pattern-making, highlighting their effect on intelligent garment engineering.

(a) AI-driven pattern parameter prediction

Traditional pattern construction has historically demanded a high level of expertise. To reduce reliance on skilled pattern makers, AI has been integrated into garment and pattern creation. ML, expert systems, and other artificial intelligence technologies laid a good foundation for pattern construction.[10] AI systems enhance decision-making and knowledge acquisition by leveraging a knowledge base and logical reasoning mechanisms that emulate human expert problem-solving, utilizing comprehensive information including body measurements, sizes, pattern information, ease allowance, and fabric mechanical properties, and then carry out the computational processes required for pattern generation.

In the domain of pattern parameter prediction, AI is used to develop predictive models, playing a crucial role in optimizing design, production, and customization processes. Fuzzy logic (FL) and simple ML algorithms, such as linear regression (LR), support vector machine (SVM) are commonly applied to analyze and interpret complex patterns in garment-related data. For example, Li et al.[69] developed a pattern dimensions prediction model based on garment flat recognition using SVM algorithms.

The application of DL in garment pattern design represents an advanced stage beyond traditional ML approaches, offering enhanced capabilities in handling complex, nonlinear relationships and large-scale data processing. Neural networks exhibit an extraordinary capability to derive meaning from complicated or imprecise data.[70] Common DL algorithms include artificial neural networks (ANNs), convolutional neural networks (CNNs), each excelling in different aspects of garment pattern design. For instance, ANN is particularly effective in nonlinear computations, enabling the prediction of garment dimensions and ease allowances based on body measurements and fabric properties.[71] Hu et al.[72] utilized a neural network combined with an immune coevolutionary algorithm to predict pants fit and determine optimal sizing, demonstrating the application of intelligent algorithms in personalized garment adaptation. CNN, known for its strength in image processing, is frequently employed to extract body measurements and predict garment specifications from 2D images, enhancing accuracy in virtual fitting and automated pattern generation.

Compared with traditional ML approaches, which primarily focus on structured data analysis, DL enables a more sophisticated, data-driven understanding of garment design by learning intricate patterns directly from raw visual and numerical inputs. This advancement facilitates greater automation, improved accuracy, and more adaptive solutions in garment pattern engineering.

(b) AI-generated pattern design (AIG-Pattern)

Generative AI has garnered considerable attention in the field of fashion design. AI tools assist in the creation, editing, and stylization of patterns. In general, different generation tasks can be differentiated by varying inputs, such as images, text prompts, point cloud data, sewing patterns, 3D models (unclothed or clothed human model, garment model, human motion in model sequences), or videos. These input modalities can be utilized independently, for instance, in text-to-3D or image-to-3D generation, or combination. The integration

of multimodal data in garment and pattern generation has become a key research focus, as it facilitates the capture of intricate details that unimodal data alone cannot provide.

Several advanced generative AI technologies are frequently utilized in the creation of garment patterns, including generative adversarial networks (GANs), variational autoencoders (VAEs), transformers, diffusion models, autoregressive models, and multimodal models. Moreover, numerous garment and pattern generation pipelines incorporate commercial AI models, such as GPT-4, DALL-E, Midjourney, and Stable Diffusion, to facilitate text and image generation. Furthermore, the integration of CLO3D and Style3D enables highly realistic and precise physical rendering of 3D models, enhancing the fidelity of digital garment simulations. For example, DressCode[73] is a text-driven 3D garment generation framework, using a GPT-based autoregressive model to achieve sewing pattern generation with text prompts and using a pretrained Stable Diffusion to generate physical-based rendering textures. Design2GarmentCode[74] integrated Style3D to visualize garment pressure. ChatGarment[75] can convert garment pictures into a 2D sewing pattern and then enable the user to use language to edit the pattern via large vision-language models.

Such AI models are not solely designed for garment pattern generation; rather, they can integrate multiple tasks, including pattern creation, texture generation, virtual try-on, and human motion synthesis. AI-generated designs assist both novices and professionals, enhancing efficiency and creativity in garment development. Moreover, these models offer greater flexibility and diversity in utilizing human body data, enabling more adaptive and personalized design solutions. To sum up, the role of AI in garment pattern design is poised to expand, fostering increased efficiency and a data-driven approach within the fashion design process. Figure 6 illustrates AI tools for garment pattern design.

Despite significant advancements in AI, its application in garment pattern design faces three major technical challenges. First, AI lacks the intuitive and emotional reasoning required to interpret creative design language. Experienced pattern makers incorporate brand identity, stylistic intent, and fabric characteristics to make informed decisions that balance aesthetics and functionality, which current AI models are unable to replicate. Second, AI struggles to accurately capture the designer's specific expectations for patterns. The implicit knowledge developed through years of practice, particularly in understanding the complex relationship between the human body and garments, cannot yet be effectively modeled. As a result, designers often need to revise and refine input prompts repeatedly. Third, AI-generated outputs are difficult to edit, refine, and apply to actual garment production.

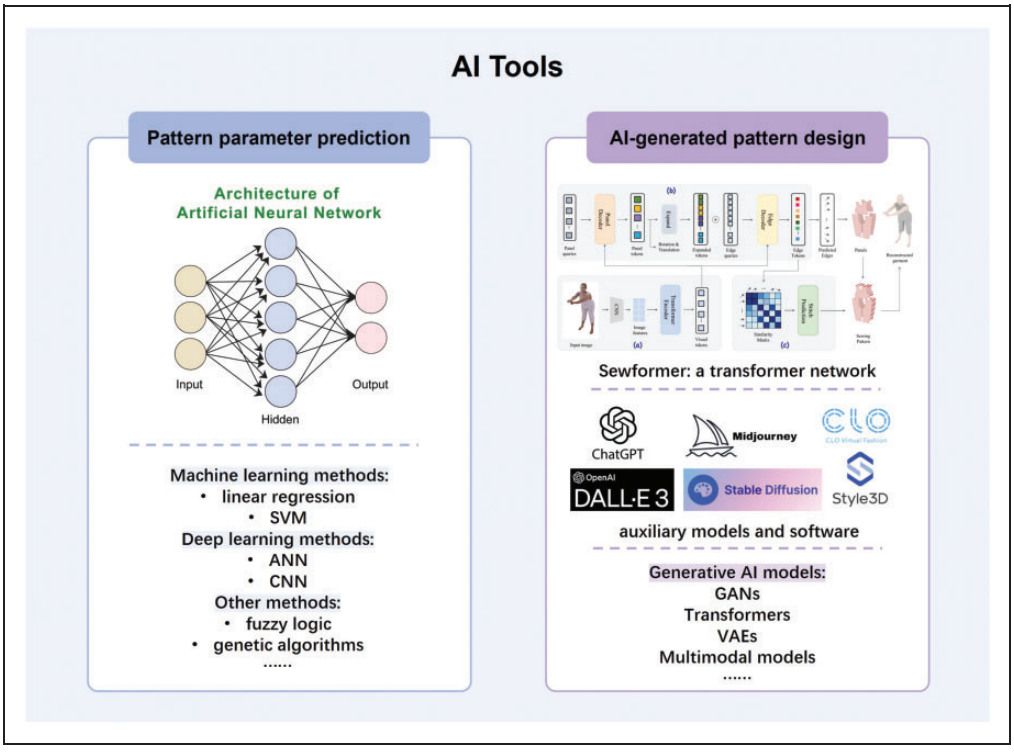


Figure 6. AI tools.[76]

These outputs typically require human evaluation and adjustment to meet the standards of fit and manufacturing quality.

Body data-driven garment pattern construction

Pattern construction methods based on 1D body data

Parametric design. Parametric design implies using parameters to define a form when what is actually in play is the use of relations.[77] Three essential components in parametric pattern design systems are provided in Table 4. The parametric pattern program automatically generates garment patterns using a predefined programmed parametric formula,[78] derived from anthropometric correlations. The generated pattern shape can be modified by changing the 1D body measurements of the target individual.

At present, there are mainly two categories of parametric garment pattern design: one using the upgraded garment-specific CAD mapping software or 3D modeling software, and the other using the secondary developed function to drive other general drawing or digital image processing software, such as MatLab data visualization function or AutoCAD parametric drawing.[81] Lee and Han[80] compared three known parametric clothing pattern CAD software, namely YUKA, GRAFIS, and Seamly2D, analyzing the types of entities, constraints, and parameters, and how they interact in each system. Table 5 lists some typical studies on parametric pattern design in recent years.

Table 4. Three essential components in parametric pattern design

Components	Description
Entities	Points, lines, arcs, curves, and others.
Constraints	Geometric constraints (incidence, coincidence, distance, angle, parallelism, concentricity, tangency, perpendicularity, and similar measurements),[79] dimension constraints.
Parameters	Body measurement, design or pattern measurement, formula parameter.[80]

The usability of a parametric pattern design system lies in its efficient integration with body measurement technology. Traditional systems require users to manually input or import explicit 1D body measurements, such as DisDraw[62] and JBlockCreator.[21] However, some advanced systems allow users to directly upload front and side photos or 3D scan files, with the system automatically extracting 1D body measurements as parameter inputs, such as the AutoCAD-based system developed by Lu et al.[16] Dāboliņa et al.[86] obtained body measurements from 3D scanning data using automated data extraction software ScanWorx and then used parametric CAD system to generate individualized patterns.

Some parametric pattern systems provide advanced functionalities, including simulation and data analysis. Kim[85] proposed a feature-based parametric pattern generation method, where once the patterns are assembled on a deformable virtual body model to form a garment, its shape and size can be easily modified by regenerating the patterns using appropriate anthropometric data. Specific systems even allow for the analysis of the relationships between pairs of measurements.[14]

In recent years, parametric pattern-making systems have not seen significant technological advancements, which can be attributed to several interconnected challenges. First, parametric design is well-suited for simple, standardized garment styles such as shirts, pants, and suits, but struggles to accommodate more complex, asymmetrical designs. Second, each unique garment style requires its parametric model, which entails a time-consuming and labor-intensive process for developers to establish specific drafting rules. This makes it challenging to keep up with the rapidly changing fashion trends. Lastly, while parametric models often rely on simplified linear formulas, the actual relationship between body measurements and garment patterns is nonlinear, further complicating the process.[84]

Table 5. Typical parametric pattern research in recent years

Study	Year	Software or language	Garment type	Curve line drawing method
Jin et al.[81]	2023	Python	Women's shirts	Curve fitting
Ye et al.[82]	2023	Python	Female prototype and dress	Biarc
Kang et al.[83]	2021	—	Female dress and pants	B-spline interpolation
Harwood et al.[21]	2020	Java	Trousers, skirts, bodices and sleeves	—
Liu et al.[84]	2019	—	Women's jeans	—
Kim et al.[85]	2012	A text-based script language	Men's top with sleeve	—
Xiu et al.[62]	2011	—	Rear panel of a women's jacket	—
Lu et al.[16]	2010	AutoCAD	Shirts, pants, skirts, and jackets	Curvature parameter

2D grading. To differentiate from the 3D grading (described later), the traditional pattern grading is described as 2D grading in this section. Grading is not a process of creating patterns "from scratch"; instead, it is creating sized patterns by increasing (or decreasing) the base size pattern by set increments and proportional rules[87] while maintaining the appearance of the garments' style. Grading is the most practical method because of its simple theory in the pattern development process.[88] It eliminates the drafting process for different sizes, and the quality of the graded patterns depends on the grading technique. The pattern maker must make several decisions about how much each pattern needs to change and exactly where each pattern needs to change.[65]

Statistical data from body measurement surveys, or the body measurements of an apparel company's fitting model, play a critical role.[89] Grading is based on size charts and is used either for standard size adjustments or calculating the specific alterations required.[12] However, it is a common complaint that traditional grading practices fit only body shapes similar to the base model. In reality, the human body does not grow proportionally, as implied by size charts, leading to grading limitations in accommodating diverse body shapes and causing fitting issues.[65,78,90,91]

Grading's made-to-measure (MTM) function in mass customization has been widely explored. Han et al.[90] integrated "automatic drafting by pattern formulae" and "grading" in their system, using body measurements, a basic size pattern, grading values, and alteration rules to generate and adjust graded patterns for individualized designs. Lim and Istook[64] developed an MTM system utilizing grading and alteration rules based on ASTM 5585 standard sizes. Han[89] examined the relationship between chest girth and height with men's upper body dimensions to develop a grading system for semicustomized clothing.

To enhance the adaptability of grading patterns to accommodate various 3D body shapes, Han et al.[78] proposed a width–height independent grading method that calculates grading values from body measurements, eliminating complex alteration rules and improving on traditional proportional grading. Moreover, researchers have incorporated 3D body shape data into grading practices. Bye et al.[91] selected Misses sizes 6–20 based on ASTM D5585 and conducted quantitative and qualitative analyses to compare traditionally graded patterns with fit-to-shape patterns. Similarly, Song and Ashdown[63] optimized trouser patterns for three distinct lower body shapes to evaluate whether grading from shape-specific blocks enhances fit. Both studies emphasized the vital role of incorporating body shape into grading practices to improve garment fit.

In summary, grading primarily relies on size charts and grading rules established from 1D body data, enabling the rapid creation of patterns for individuals of various body sizes and types, which holds significant potential for mass customization. Despite the optimization of grading methods for pattern personalization, certain limitations remain. First, the wide variation in garment styles makes it impractical to predefine universal grading and alteration rules.[92] Second, grading also depends on the expertise of pattern makers, varying by garment type, such as suits and shirts.[14] As garment complexity increases, alteration rules become progressively more intricate.[65]

Graphic flattening. This section summarizes studies that begin with an anatomical perspective but construct patterns with measurements. Nakazawa[93] conducted extensive research on the relationship between body anatomy and garment pattern construction and further constructed the prototype pattern for the upper body by spreading the bust line, shoulder line, and armhole curve, taking into account the distribution of ease allowance, laying a foundation for subsequent studies. Gu et al.[94,95] and Su et al.[96,97] established a connection between measurements extracted from frontal and side views of 3D scan images and corresponding prototype pattern dimensions, thereby facilitating the individualized pattern design of suits and pants. Similarly, Kim and Choi[98] utilized leg measurements extracted from body scans to create a legging block pattern. The pattern incorporated horizontal and vertical baselines, defined based on landmark points, primary levels, and standardized body measurements. The principle of converting surface length measurements into a 2D pattern involved rotating the horizontal line around the vertical sideline to align the surface length and curving the outer line accordingly. These methods convert 3D body data into 1D measurements, reducing the complexity of pattern design with 3D data but at the expense of body information details. Graphic flattening, while less convenient than parametric and grading methods, also falls short in capturing the comprehensive body data representation achieved through 3D pattern-making. As a result, it is primarily used in academic research to illustrate the transformation between 2D patterns and 3D bodies, but has yet to achieve widespread commercial adoption.

AI-aided flat pattern-making. AI has been integrated into pattern design for an extended period. ANN and FL were among the first algorithms introduced for generating garment patterns. ANN is a nonlinear data modeling system that can establish models or patterns in complicated relationships between inputs and outputs,[70] facilitating improved performance over time

and the ability to apply knowledge to new, unseen scenarios in garment pattern generation. Conversely, FL effectively addresses vague, ambiguous, or imprecise issues, such as those related to garment fit or ease. The utilization of both ANN and FL aims to identify parameters that define the topological relationships between points, lines, and curves in pattern construction, thereby enhancing the precision and adaptability of the process.[97]

Chan et al.[99] predicted shirt pattern sizes for different body sizes using multiple linear regression (MLR). The underlying relationship between shirt pattern parameters and body measurements, as examined through MLR, was found to be more accurate than the formulae provided by the pattern expert. Traditional drafting formulas rely on oversimplified linear relationships, whereas nonlinear systems, such as those utilizing ANN, can handle more complex and intricate relationships in pattern-making. Huang et al.[100] employed a back-propagation (BP) ANN to predict the anthropometric data for women's upper tops pattern-making. Ma and Yan[10] developed an intelligent pattern design system based on BP neural networks, utilizing ANN's self-learning, self-organizing, and nonlinear mapping functions.

Chen et al.[101] employed FL and sensory evaluation to optimize the estimation of garment ease allowance, effectively modeling the relationship between ease allowance and various factors such as body measurements, garment technical parameters, and comfort requirements. Inui[102] utilized a genetic algorithm to code sleeve pattern dimensions and associated calculation parameters. Wang et al.[103] developed a knowledge base using sensory analysis, ontology, and support vector regression modeling. This system learns the quantitative relationships between garment structure lines, control points, and patterns, enabling the prediction of pattern parameters and facilitating adaptive pattern design.

Pattern construction methods based on 2D body data

Ease allowance design. In general, a gap exists between the body and the garment, referred to as ease allowance. It can be classified into radial ease allowance and girth ease allowance in the literature.[5] Ease allowance determines the functionality, comfort, fit, design, and style of the final garment.[8] Establishing the ease relationship between the body and the garment is essential. Traditionally, pattern-making methods use some direct additions of ease amounts to body measurements,[104] yet they fail to provide accurate results when applied to 2D-to-3D spatial transitions. With the obtained body cross-section information, the ease allowance distribution can be visualized in pattern construction.

Wang et al.[5,105] developed a mathematical model to represent the ease distribution of cross-sections at different altitudes for an X-line style jacket, applying predicted segmental girth ease to pattern alterations. Similarly, Su et al.[106] and Gu et al.[107] investigated the distribution of distance ease across essential transverse planes in pants and suits to facilitate pattern alterations. In bifurcated garments, the U-shaped crotch curve in the sagittal plane plays a pivotal role in achieving pants fit. Gu et al.[108] developed a predictive model correlating the distribution of distance ease along the crotch curve with the crotch ease allowance. Then, the predicted distance ease is applicable when adjusting patterns to enhance the fit along the crotch curve.

Some studies have explored cross-section ease allowance in 3D space. Tao and Bruniaux[109] defined an ease model of leg cross-section contours to drape a fit pant model in CAD software. Similarly, Hlaing et al.[110] prototyped a parametric virtual pant model by scaling the female lower body contours. Wang et al.[20] created a set of horizontal planes on a parametric body model, then distributed ease allowance into cross-section planes to generate developable garments. The garment surface was converted into 2D flat patterns following a flattening process. Nevertheless, the enlargement of the intersection curves relies on ease allowance values derived from traditional 2D pattern-making techniques, which do not yield accurate 3D spatial relationships. Presently, data on the ease distance between the body and the garment in 3D space remain relatively limited. Reverse engineering offers a robust approach to calculating ease between scanned garments and the body, the measured ease values then serve as a reference for designing virtual ease allowances in a 3D environment. Thomassey and Bruniaux[61] developed an ease allowance template to design basic men's bodice and sleeve patterns using a reverse methodology, thus creating garment patterns containing real ease allowance amounts. In summary, the ease distribution along cross-sectional curves derived from transverse or sagittal planes is crucial for accurate pattern construction, the added ease amounts influence the final pattern shape.

Sketch modeling methods. Sketch modeling uses sketched 2D garment contours to generate a 3D garment model, even developable patterns.[111] At present, the main functions of sketch modeling systems include user interactions, 2D geometry creation, cropping and extension, sketch constraints, and sketch file storage and retrieval. Most garment sketch modeling systems deal with silhouette curves, cross-sectional contour curves, and style curves (such as darts, creases, and

inner cutting lines). Users can control and edit garment contours in a 2D view by referencing body cross-sectional curves or silhouette curves to achieve the desired garment design, they can even specify certain pattern details to refine garment construction. The key is to leverage the spatial relationship between the human body model and the sketched 2D garment silhouette to infer distance variations in three dimensions.[112] Sketch modeling systems are inherently visual and interactive, aligning with designers' working practices, as most still prefer to express their creative ideas through 2D sketches.[113]

In previous studies, most research focused on generating garments without further developing patterns.[112,114–117] Turquin et al.[116] proposed a straightforward method that allowed users to sketch garment contours directly onto a 2D mannequin view, using a distance field to compute the 3D garment surface. Later, Turquin et al.[112] incorporated the drape effect of actual fabric into the system by allowing strokes to represent ridges or valleys. Decaudin et al.[118] further improved Turquin et al.'s system by adopting a fully geometric approach, enabling users to sketch additional seam lines directly on the garment surface, with 2D flat panels flattened through the ABF++ parameterization method, as shown in Figure 7(a). These studies are more suitable for computer animation due to the lack of practical pattern construction considerations.

To be more practical, Wang et al.[113] developed a template based on human body features and utilized a modified variational subdivision scheme to generate mesh surfaces. The 3D garment profiles were defined by 2D sketches input, enabling the design of loose garments by incorporating the easing space between the garment profiles and the body contours. Zhang et al.[111,119] interactively generated garment surfaces using a sketch-based modeling method and modified the garment shape through control curves based on the silhouette and cross-section lines of the human model, as shown in Figure 7(b). They further developed 2D patterns by flattening the 3D surface patches. Wang et al.[120] used parametric curves to interactively design 3D garments and flattened patterns through the mass–spring model.

AI-aided image-based pattern generation. Body images were initially used for body measurements and garment size prediction through AI techniques. The data obtained from these predictions were then applied to flat pattern making. For example, Jeong et al.[121] utilized predicted pattern parameters from a single garment photograph to drive a parameterized pattern drafting module, enabling the creation of garment panels and virtual garments.

As AI capabilities continue to evolve, using 2D images as input for generating virtual garments and sewing patterns has attracted significant interest in the computer graphics community. Researchers have employed various techniques to extract features from garment images, predict 3D point clouds, and reconstruct sewing patterns based on these predicted 3D models. In this process, large datasets comprising garment models, images, patterns, and bodies are essential, while fine-tuning the model architecture plays a crucial role in ensuring successful implementation. For example, Li et al.[122] used differentiable simulation to optimize 2D patterns, allowing the recovery of clothing assets from 3D scan images. Lim et al.[123] focused on reconstructing 3D garment models from a single image of a posed user, estimating sewing pattern parameters based on an inferred T-pose garment image to prevent unnatural garment appearances, as shown in Figure 8(a).

Currently, Transformers and GANs have increasingly emerged as mainstream generative neural network architectures. Chen et al.[124] introduced a transformer-based deep neural network Panelformer for reconstructing complete sewing patterns from garment images. AIpparel[125] leverages multimodal inputs, including images and text, to generate intricate, diverse, and high-quality sewing patterns, thereby enabling

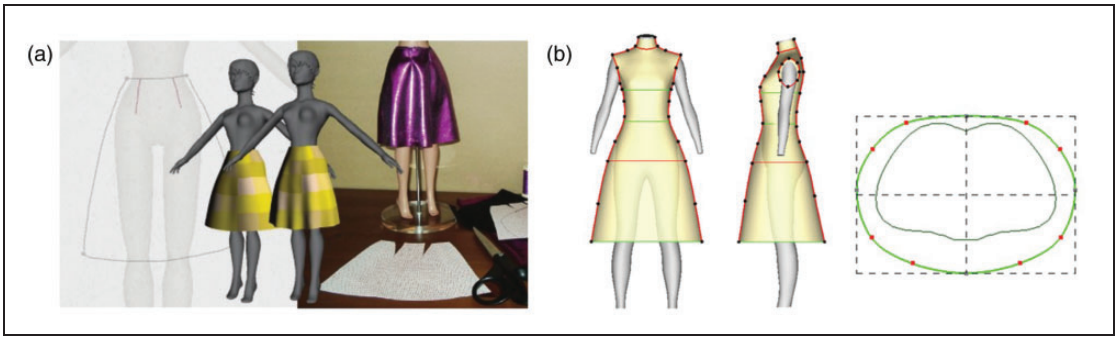


Figure 7. Sketch modeling: (a) from sketched contours to garment[118] and (b) modify 3D garment shape by control curves.[119]

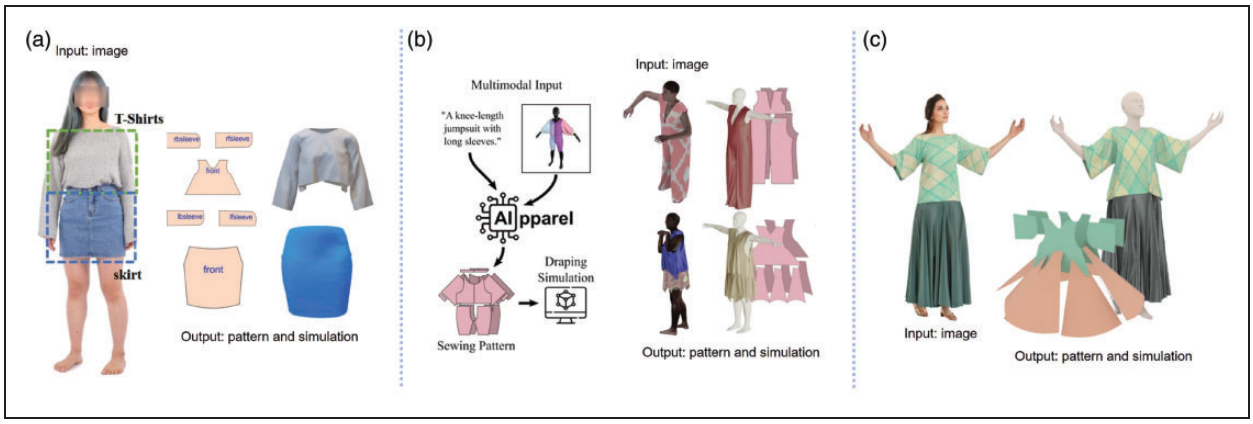


Figure 8. AI-aided image-based pattern generation models: (a) SPnet[123]; (b) AIpparel[125] and (c) Sewformer.[76]

novel applications such as language-guided sewing pattern modification, as illustrated in Figure 8(b). Sewformer[76] is a two-level Transformer network designed to recover garment sewing patterns from daily photos, significantly enhancing pattern prediction accuracy, as in shown Figure 8(c). Shen et al.[126] proposed a GAN-based generative model capable of producing diverse garment meshes based on input sewing pattern images, body meshes, and ultraviolet (UV) maps. Despite their convenience in the design process, these methods produce patterns that lack structural elements such as darts, limiting their applicability in practical garment manufacturing.

Pattern construction methods based on 3D body data

Body surface flattening. Surface flattening can be defined as the process of determining a 2D pattern from a given 3D freeform surface and establishing a mapping between them.[127] Commonly used flattening algorithms include ARAP (As-Rigid-As-Possible) parameterization, ABF++ (Angle-Based Flattening), physical spring–mass models, and finite-element methods (FEMs). The flattening process must follow key principles, aiming to minimize distortions in angles, area, or length between the 2D and 3D meshes.[128] Color gradients enable computer-aided tools to visualize stretching within flattened patterns.[14] Inevitably, the flattened pattern undergoes some corrections of construction lines and darts position.[8]

3D scanning data greatly improves the clothing fitness. For specific tight-fit functional garments or sportswear with little or negative ease allowance, the pattern can be developed directly from 3D human body or mannequins by flattening since the 3D garment patterns take up the contours of the body,[129] for example, compression garments,[130] cycle wear,[131] garment for atypical bodies,[23,132] female bodice, bra,[133] and life jacket.[134]

3D human body data can be used to create a subsurface for flattening and, thus, impart the original anthropometric features of the input human model to the flattened pattern. Huang et al.[135] used scanned 3D unstructured points of the human body to generate block patterns through a series of steps, including body recognition, body triangulation, body partitioning, planar patch fitting, and patch coplanarization. Kim and Park[136] generated basic patterns using a 3D geometric modeling method, where the fit zone derived from 3D body scan data can be resized parametrically, and the garment model can be projected into corresponding flat panels. Kim and Kang[137] first generated a body model using 3D body scan data and a garment model through the stereovision technique, then used a surface warping algorithm and optimum planar pattern mapping algorithm to generate 2D garment patterns.

Scanned mannequin data can also be the source for flattening, and the corresponding flattened patterns are more regular than directly flattening the human body surface. Yang and Zhang[128] segmented a 3D virtual dummy into 10 regions according to the Bunka garment block, then generated prototype pattern through flattening. However, the 10 patches tend to ignore the deformation features of the human body.[135] Kim et al.[138] reconstructed a body surface model from 3D mannequin data, adjusting the size and shape of the mannequin model using free-form deformation (FFD) to match the human subjects' bodies. The flat garment patterns, incorporating multiple darts, were then generated through the flattening process. Kang and Kim[139] generated a convex-shaped dummy model using 3D anthropometric body data and defined the garment model using stereoscopy, finally flattening the garment surface into 2D patterns. To sum up, the drawbacks of the surface flattening technique are evident, including distortions and issues with ease

addition, making it challenging to apply in actual garment production.

3D grading. The concept of 3D grading of patterns is realized by constructing the same garment on differently sized virtual human models.[13] If the mapping between the new garment and the reference garment is established, pattern grading can be easily achieved.[119] The automatic resizing function has become a critical component of 3D garment CAD systems, enabling garment generation tailored to individuals with diverse body shapes and measurements.[9]

One commonly used method for 3D grading is garment transfer. It is based on deformation transfer, which applies the deformation exhibited by a source triangle mesh onto a different target triangle mesh.[140] Wang et al.[141] proposed a computer-graphics-based approach for developing an automatic MTM system, which enables designing garments on one 3D body and rearranging them to different morphologies. Wang et al.[142] presented a volume parameterization technique that achieves similar functionality. Brouet et al.[143] proposed a fully automatic garment transfer algorithm, preserving the design, proportionality, and fit, as shown in Figure 9(a). Considering that unwanted garment shape distortion may occur during the transfer process, Meng et al.[9] provided a shape control technique for automatically resizing virtual garments through shape optimization and garment surface deformation. Following the surface flattening procedure, all these 3D grading methods facilitate the generation of corresponding 2D garment patterns, enabling accurate translation from 3D models to 2D designs.

There are also other methods that achieve 3D grading effects. Chen et al.[144] utilized a differentiable simulator to drape a 3D garment onto bodies of various sizes and adjust the corresponding 2D sewing pattern, as shown in Figure 9(b). Korosteleva and Sorkine-Hornung[145] implemented a body retargeting method in which garment components are parameterized by body measurements, making it easier to fit a garment design on different bodies. The sketch-based methods[111,113] mentioned earlier also incorporated a 3D grading function by automatically refitting garments to bodies of varying sizes and shapes. Wang and Huang[20] employed a parametric human body model, deforming it based on an industrial sizing chart to generate a range of well-fitted and flattenable 3D garments, as shown in Figure 9(c).

Virtual draping. Virtual draping employs a method similar to real draping in virtual space. The implementation of virtual draping has been made possible by advancements in computer hardware performance and the development of sophisticated drape simulation technologies. The primary advantage lies in enabling designers to interactively and visually manipulate virtual garment designs while enhancing efficiency through the streamlined digital workflow. Cho et al.[146] used computed geometrical models to drape the skirt panels based on a 3D dummy, then converted 3D contoured panels into 2D patterns. Sul and Kang[147] modeled fabric as a virtual rectangular sheet draped over a mannequin, with redundant sections removed via a virtual scissoring tool conducting mesh-based operations to prepare the initial pattern shape in 3D space. The system generated 2D flat patterns by replicating the scissoring process, bypassing traditional surface flattening techniques. Mesuda et al.[148] proposed a virtual draping method for pattern generation, wherein a rectangular planar cloth model was created and modified with necessary slits for darts. The altered model was then mapped onto the surface of a mannequin to obtain a 3D pattern, from which excess material was removed. Subsequently, the 3D pattern was inversely mapped back onto a plane to derive the final planar pattern. Fang and Ding[149] employed virtual scissoring to design a 3D garment on a reconstructed digital human body, then flattened the garment surface into 2D basic patterns.

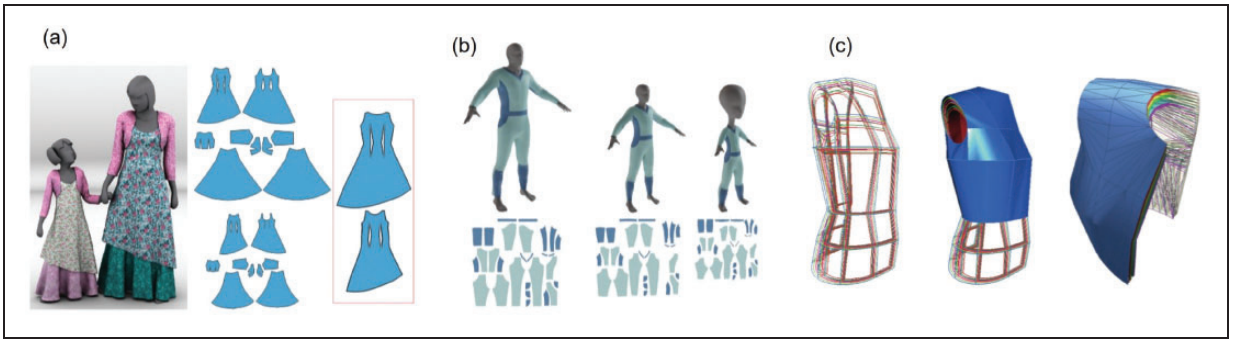


Figure 9. 3D grading methods: (a) design preserving garment transfer[143]; (b) Dress Anyone, physically-based garment pattern refitting[144] and (c) 3D graded frames and garments.[20]

To explore the potential of augmented reality (AR) and virtual reality (VR) in virtual draping, Inui et al.[150] developed a virtual draping system using AR by modeling cloth, hand, and dress form components. The cloth's dynamic shape was mechanically simulated, while hand motions were detected via sensors and reflected in a virtual environment. The system enabled users to manipulate virtual fabric through real-time hand gestures, integrating virtual and real-world images to enhance usability. The intuitive 3D design platform Gravity Sketch offers a digital draping method powered by its advanced 3D geometry engine, which enables seamless sketching, volumizing, and surfacing in Subdivision Surface, nonuniform rational B-splines (NURBS), or mesh, facilitating efficient 3D content creation and export. The draping process begins with taping the mannequin with key landmark lines, such as princess lines, serving as draping guides. Muslin fabric is then virtually pinned onto the mannequin, allowing for precise digital draping across the body, as shown in Figure 10(a). Expanding its exploration of virtual draping in VR, Gravity Sketch has collaborated with PUMA, as shown in Figure 10(b), enabling designers to immerse themselves in creative ideation, freely modify styles and colors, accelerate design iterations, and receive real-time feedback from colleagues.

Although the virtual draping method is visually intuitive and enables rapid prototyping, it is constrained by its applicability to simple designs and poses challenges with complex garment structures. In addition, its efficiency is not guaranteed due to the iterative cutting procedures and the requirement for specialized expertise.

AI-aided scan-based pattern generation. Recent advancements in AI-aided scan-based pattern generation have significantly streamlined the design process by converting 3D body scans into 2D sewing patterns.

The acquisition of highly detailed 3D clothed human body is becoming increasingly accessible. However, scanned clothed body data lacks essential information, such as the segmentation between clothing and body, as well as garment pattern information,[151] thus limiting its broader application. To address the practical need for garment patterns in manufacturing, many studies have been conducted to generate sewing patterns from 3D clothed scans. For example, Goto and Umetani[151] estimated garment patterns from scanned individuals in clothing using DL to derive patterns from multiview projected images. Bang et al.[152] estimated garment patterns from static scan data by first segmenting the garments using an implicit boundary representation and an optimization strategy, the segmented garment pieces were then flattened into 2D patterns.

Compared with scan data of a clothed human body, scan data containing only the human body or only the garment is easier to use in pattern generation tasks, as it eliminates the need for segmentation. For example, Danner et al.[153] introduced an AI-assisted pattern generator, this tool automates the transformation of 3D body scans into flattened 2D patterns, accommodating various garment styles, from close-fitting to loose-fitting designs, by adapting to individual body measurements. The integration of ML algorithms allowed for customization and broad applicability across different designs. Korosteleva and Lee[154] developed a novel architecture, NeuralTailor, based on point-level attention for set regression, which reconstructed 2D garment sewing patterns from 3D garment point clouds. Experimental results demonstrated its ability to generalize to unseen garment types, offering a unified approach beyond specialized models. These sewing

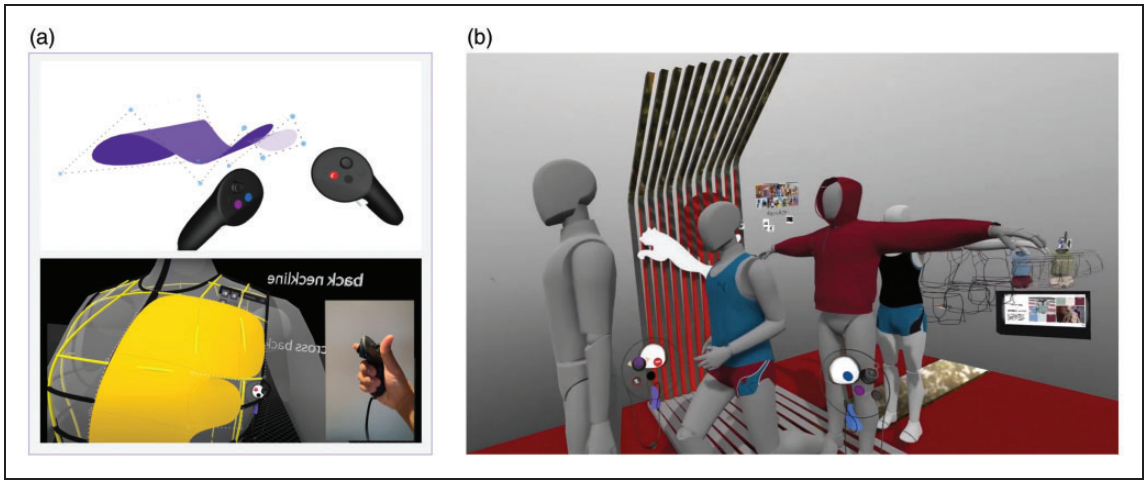


Figure 10. Virtual draping in VR: (a) Gravity Sketch digital draping method and (b) Gravity Sketch and PUMA collaboration (https://gravitysketch.com).

patterns provide essential details, including 2D-panel shapes and their relative positions to a reference body.

Although these methods utilized 3D body data to generate 2D patterns, they lacked sufficient integration of garment pattern expertise, resulting in various imperfections. For instance, the generated patterns exhibited coarse lines, insufficient curve smoothness, and misaligned seam junctions between different pattern pieces, ultimately failing to meet manufacturing standards.

Pattern construction methods based on dynamic body data

Pattern-making based on dynamic measurements. In contrast to traditional flat pattern-making methods that rely solely on static body measurements, using body dimensions from dynamic postures provides a more ergonomically informed approach. This method initially uses static body measurements and patterns as a baseline, followed by an analysis of movement characteristics to integrate dynamic body changes, such as joint range of motion and skin extension, into the static pattern. This approach enables the development of principles for constructing dynamic patterns.

Wang et al.[4] analyzed measurement differences in Chinese males between static and dynamic states and subsequently developed dynamic block patterns for shirts and trousers. Bogović et al.[155] conducted dynamic measurements of sports aircraft pilots to develop protective clothing patterns, emphasizing the sitting posture within the cockpit. Avadanei et al.[156] considered dynamic changes in back length during bending postures to improve the design of overall protective garment patterns. Shi et al.[157] investigated skin deformation in cyclists using a motion capture system during dynamic cycling in a bent posture, the mean skin deformation rates derived from this study were applied to adjust the patterns of compression cycling skinsuits, enhancing aerodynamic properties and supporting improved muscle movement.

Some researchers used nonwoven or nonstretch woven fabric to construct a static garment prototype, analyzing the locations and extent of tears or distortions in the garment after wear, this information was subsequently used to add additional ease allowance to the static pattern. Simões[158] analyzed the after-wear distorted contour of the garment made of non-stretch woven fabrics, then made alterations to basic patterns for mobile bodies. Bragança et al.[159] designed a dynamic work-wear pattern to accommodate various working postures, utilizing nonwoven garment tear and compression tests.

The primary limitation of these methods is that the measurement of crucial body postures within a movement sequence remains essentially static, failing to capture the continuous changes that occur during human motion. Moreover, the muscle state of the body in a static posture inherently differs from that in active motion, and the influence of this discrepancy on body measurements has yet to be thoroughly investigated.

Flattening of 3D dynamic body models. The dynamic human body flattening method does not strictly depend on 4D data; instead, it utilizes discrete sequences of 3D models derived from 4D frameworks. This method, based on principles of human anatomy and movement morphology, utilizes digital dynamic models to depict distinctive body postures during specific activities. Segmenting lines are designed on the body as garment construction lines, typically referencing natural, non-extension skin lines, followed by a flattening process.

For example, Xiao et al.[160] utilized a mass–spring physical model to generate flattened 2D pattern from deformed human body models with different skating poses. Jolly et al.[161] employed kinematic analysis to identify the typical poses of motorcyclists, subsequently constructing posture-based body models using the SMPL parametric model and ultimately unwrapping the 2D pattern pieces from dynamic models. Wolff et al.[162] directly created 3D garment shapes on a range of 3D-scanned bodies with poses based on SMPL and obtained 2D patterns through parameterization, ensuring both dynamic fit and the desired appearance.

Based on a similar consideration of body dynamics, Lindqvist[163] proposed a kinetic garment construction method utilizing traditional draping techniques emphasizing biomechanical points of the body, effectively balancing ergonomic requirements and fabric properties.

Current 4D data research in fashion. The 4D technology provides researchers with a forward-looking approach to analyze, process, and visualize the morphology of a moving human body. At present, 4D data have not been applied directly to clothing pattern construction, primarily focused on the continuous changes in human body dimensions and the study of localized morphological changes. Kyosev et al.[52] analyzed length variations in the left back leg during motion using the homologous mesh vertices information captured by 4D technology, as illustrated in Figure 11(a). This qualitative analysis can be applied to the design and dynamic fit evaluation of high-performance sportswear. Pei et al.[53,164] employed 4D scanning technology to measure breast shape changes and vertical displacement during running. Uriel et al.[165] used a 4D scanner to obtain continuous body scan sequences during four

movements: running, vertical jump, trunk flexion with feet touching, and squatting. They developed anthropometric definitions for these dynamic states, observing the cyclic variation in measurements across each movement cycle, as shown in Figure 11(b). Klepser et al.[166] investigated the fit in motion using a 4D photogrammetry scanner system and established an assessment matrix measuring the distance from the waist point to the waistband during leg flexion. All the acquired 4D body data are product-oriented; however, their application in pattern construction remains largely unexplored.

In the field of knitted garment design, Liu et al.[167] introduced a computational pipeline for designing and fabricating 4D knitwear with optimized elasticity for body motion, utilizing a physics-driven method to compute elasticity distribution and translate it into a knitting map for digital knitting machines. Due to the flexible fabric performance of knitted garment design, which allows for close conformity to the human body and does not strictly rely on the cutting and sewing process, integrating knitting technology with 4D body data presents a highly promising approach for exploring the application of 4D data in garment design.

Recently, some scholars have attempted to apply 4D scanning data to the design of functional garment patterns. Sadretdinova et al.[168] proposed an ergonomically balanced pattern construction method for formal jackets aimed at enhancing comfort and fit for deaf individuals. Dynamic body measurements were obtained through 4D scans capturing typical sign language movements. Initially, the researchers experimented with using maximum dynamic deviations in body dimensions during extreme postures as direct parameters for pattern design. However, this approach proved unsuitable, as it led to significant pattern distortion in regions with large motion amplitudes, compromising static appearance and potentially causing fit issues. Consequently, a multifactorial method was adopted, incorporating both material stretch properties and pattern ease allowances. Stress maps generated via CLO3D were employed to optimize material and pattern parameters, enabling quantitative evaluation of dynamic fit, as shown in Figure 12. Similarly, Peev and Kyosev[169] utilized 4D scanning to capture the initial and final phases of surfing movements. The deformation of wetsuits on the legs was analyzed using the FEM, and analogous tests were suggested for other postured body models. This study highlights the potential of FEM in exploring the impact of precise fabric mechanical properties on garments fitted to dynamic postures.

Comparisons and applications

Comparisons

Discussion of 1D–4D data utilization. The following sections provide a detailed discussion of the

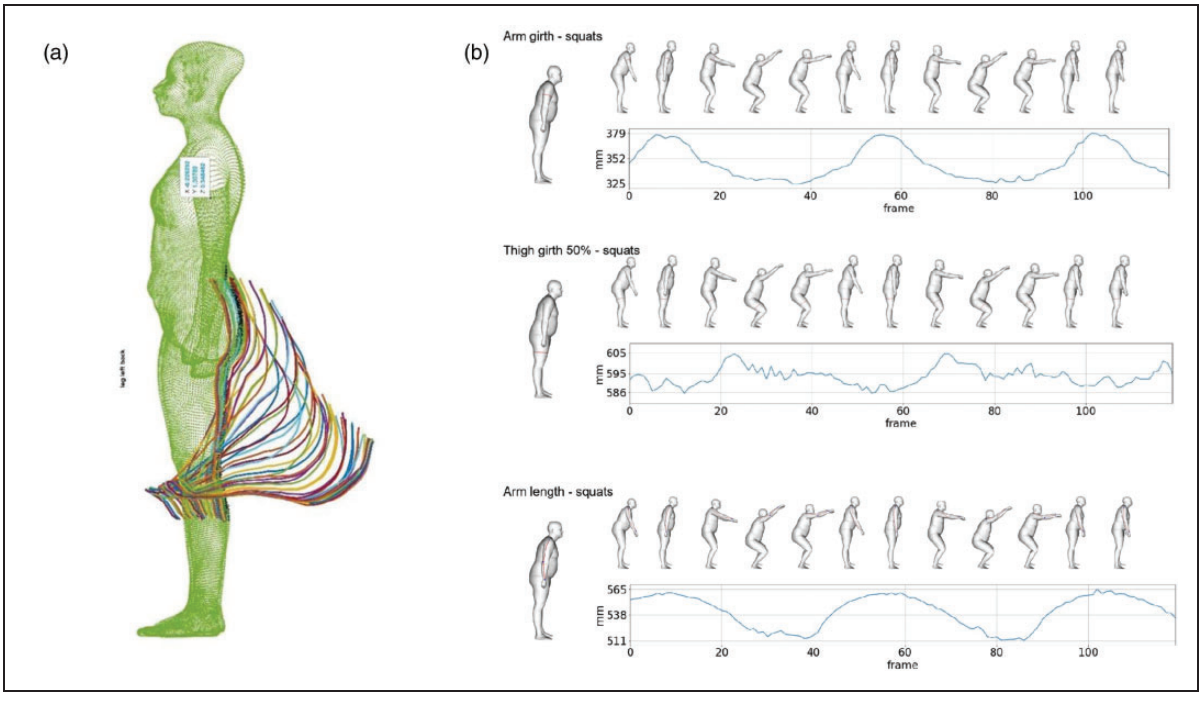


Figure 11. 4D data utilization in fashion: (a) length variations in the left back leg during motion[52] and (b) body dimension changes during squats movement.[165]

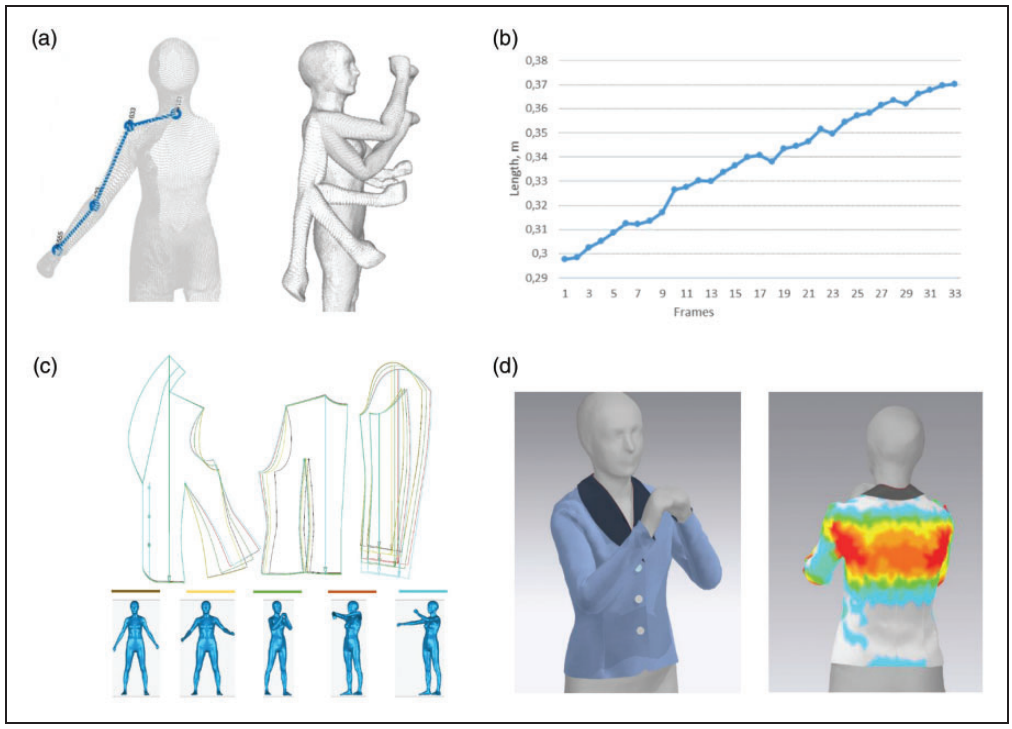


Figure 12. Dynamic approach in garment pattern design for deaf people[168]: (a) sign language movement frames; (b) length of the particular body segment during motion; (c) motion-oriented pattern design and (d) dynamic fit analysis.

advantages, disadvantages, and research gaps in using 1D, 2D, 3D, and 4D human body data for pattern construction.

(a) 1D body data utilization

The advantage of 1D data pattern construction is that it fully leverages historical pattern-making practices, making it the most aligned with industrial garment pattern-making among the four data application methods discussed in this study. With a low entry threshold and ease of use, 1D body data requires only basic dimensional measurements from the target group to drive pattern formulas that transform accurate and scalable garment blocks. Parametric and personalized grading are widely applied in both custom and mass customization fields. While graphic flattening primarily appears in academic research without practical production applications, AI-assisted flat pattern-making utilizes AI algorithms to quickly calculate necessary pattern data, thereby facilitating rapid pattern design by skilled pattern makers.

The disadvantage of 1D data pattern construction is that parametric or grading systems cannot automatically generate basic blocks from scratch, as the creation of initial block patterns still relies on skilled pattern makers. Furthermore, design variations are limited, often confined to uniform-style garments, making it challenging to keep pace with fashion trends.

The research gap in 1D data pattern construction is that current methods primarily focus on body measurement data or corresponding garment specification data, with insufficient consideration of ease allowance and fabric properties. Future research should approach 1D pattern-making from a functional perspective, investigating how to optimize ease allowance and fabric performance for garments matching specific 1D measurements. This would enable garments to achieve not only dimensional accuracy but also a comprehensive balance of fit, mobility, and comfort.

(b) 2D body data utilization

Given the significant differences among the pattern construction methods based on 2D human body data—ease allowance design, sketch modeling methods, and AI-aided image-based pattern generation—this section systematically discusses the advantages, disadvantages, and research gaps of each approach.

The primary advantage of the ease allowance design method based on cross-sections is that it enables users to understand the relationship between garment patterns and the human body from transverse and longitudinal perspectives. For instance, it facilitates the design of mobility allowances for key areas such as girth measurements, crotch, and armholes, providing detailed insights into garment fit from a sectional viewpoint.

However, due to its high operational complexity, this method remains confined to research applications and has yet to achieve practical implementation. Regarding research gaps, most studies employing this method validate garments using calico fabric. Since the distribution of sectional ease allowance is closely related to fabric properties, conclusions derived from such research may not be directly applicable to garments made from fabrics with significantly different material characteristics.

The advantage of sketch modeling methods lies in their simplicity and intuitive nature, as they do not require precise technical drawing but instead rely on rough 2D contour sketches from front or side views of the human body. However, a major drawback is the inability to obtain accurate Z-direction information during the transition from 2D to 3D, as the design is constrained to front or side views, resulting in the omission of many garment details. In terms of research gaps, with advancements in AI-driven methods within computer graphics, data-driven and ML approaches are becoming the future direction of sketch modeling techniques. Nonetheless, AI-based sketch modeling remains underexplored in the field of garment pattern design.

The AI-aided image-based pattern generation method has the advantage that 2D images are easier to acquire, and image data can be flexibly applied—not only to extract 1D human body measurements for pattern-making but also to reconstruct 3D body data for 3D pattern development. However, a major drawback of this approach is its heavy reliance on empirical estimations, lacking ground truth body data for reference. As a result, applying such methods to real human bodies can lead to significant deviations in pattern accuracy. Regarding research gaps, existing studies on garment datasets and pattern generation primarily focus on ensuring a smooth workflow and fostering creative outputs, without incorporating systematic considerations from industrial practice, such as seasonality, collection coherence, and stylistic consistency. Consequently, the generated patterns fail to establish a stable system with consistent styles and sizing characteristics.

(c) 3D body data utilization

The advantage of 3D body data pattern construction lies in its ability to achieve personalization, visualization, rapid modeling, and multiobjective generation. Firstly, personalization is enhanced through surface flattening techniques that are particularly suitable for designing compression garments, adaptive clothing, and customized apparel for individuals with atypical body shapes or physical deformities, such as scoliosis patients. Secondly, visualization is significantly improved by utilizing 3D software as a powerful visualization tool, enabling researchers to digitally create, edit, and visualize patterns in the 3D environment. This approach allows for virtual experimentation with different fabrics and patterns, facilitating seamless coordination between 2D and 3D patterns while offering suggestions for pattern alterations. Consequently, designers can achieve more accurate and visually refined designs. Thirdly, rapid modeling capabilities enable designers to quickly iterate on designs, test flat pattern fits, and make faster alterations, particularly for patterns requiring intermediate processing, such as pleats and shirring, contributing to both efficiency and sustainability. Moreover, the outstanding advantage of 3D grading lies in its ability to generate scalable garment models that maintain proportionality and detail across various sizes, which is crucial for mass production. Finally, multiobjective generation with AI tools based on large datasets enables the simultaneous creation of virtual garments, patterns, and textures. This approach is not only fast and personalized but also applicable to a wide range of fields, including AR and VR, animation, and gaming.

The disadvantage of 3D body data pattern construction is the gap between innovative computational design methods and practical industry applications, making it challenging to achieve mass production readiness. For instance, the shape of flattened patterns often requires manual post-processing, and most 3D CAD pattern construction and grading systems are primarily used for 3D garment simulations and visual demonstrations rather than actual manufacturing. Furthermore, AI-based 3D pattern generation methods often rely on pattern datasets that differ significantly from industrial patterns, making them unsuitable for direct garment production.

In addition, the transition from conventional flat CAD pattern construction to advanced 3D methods poses a challenge for garment professionals. Mastering 3D virtual draping demands exceptional computational skills, which many experienced industrial pattern makers lack. Moreover, professionals accustomed to evaluating physical prototypes may find it difficult to assess virtual garments accurately, potentially leading to incorrect judgments about product marketability and missed opportunities for bestselling items. As a result, apparel companies often continue to produce physical samples before finalizing designs. Addressing these challenges is essential for advancing the practical implementation of 3D data pattern-making in the industry.

(d) 4D body data utilization

The advantage of 4D data pattern construction lies in its ability to incorporate dynamic human factors,

which significantly distinguishes it from traditional garment pattern construction practices. By leveraging 4D body data, garment patterns are designed to align with the dynamic deformation of human skin, considering garment dimensions, fabric properties, and construction techniques. Particular attention is given to areas prone to significant skin stretch and extensive movement, such as the knees, elbows, thighs, and back, reflecting a thorough consideration of the garment's functional performance during motion.

The disadvantage of 4D data pattern construction is the high cost associated with various 4D data acquisition systems, which necessitates substantial investment. This financial burden poses a significant challenge for small and medium-sized companies, preventing them from adopting advanced 4D equipment. Moreover, the difficulty in obtaining 4D data becomes a critical barrier to practical exploration and implementation in the apparel industry.

The research gap in 4D data pattern construction is that current studies predominantly focus on 4D body data acquisition, AI-based dynamic virtual try-ons, or garment ergonomic evaluation, rather than direct application to pattern design. Given the relatively late development of 4D research and the inherent difficulty in acquiring 4D data, no mature knowledge system currently guides 4D pattern-making practices. For instance, there is a lack of scientific guidance on selecting key frames from 4D data as references for garment pattern design, and numerous technical challenges persist in addressing such research details.

Discussion of pattern accuracy. The accuracy of garment patterns is a critical concern in both academia and industry. Patterns must be checked for accuracy before production to prevent cumulative errors and avoid costly mistakes. In the digital era, pattern accuracy has significantly improved, driven by advancements in human body data precision and the development of emerging digital tools such as CAD and AI. Broadly, the evaluation of pattern accuracy can be categorized into subjective and objective assessments. Subjective evaluation, which involves rating scales completed by apparel experts, designers, and fitting models during trial fittings, is highly influenced by individual preferences, leading to limited comparability of results. Therefore, this study does not focus on subjective evaluation but instead explores objective assessment methods related to pattern accuracy.

(a) Pattern accuracy related to 1D data

Pattern-making methods based on 1D body measurements, such as parametric design and grading, are widely used in the apparel industry and require a high level of accuracy to ensure responsible garment design and manufacturing. In such pattern-making practices, the greater the number of 1D measurements used, the more closely the pattern aligns with body shape information, resulting in a better fit. For instance, the accuracy of patterns increases progressively across three production methods: ready-to-wear (RTW), MTM, and bespoke. RTW aims to use a limited set of basic 1D measurements to achieve maximum population coverage, aligning with the goal of size inclusivity. In contrast, bespoke tailoring, such as custom suits, incorporates dozens of body measurements to achieve a highly precise and personalized fit.

The evaluation of pattern accuracy can be considered from two perspectives: the flat pattern and the garment sample. At the flat pattern level, a key assessment metric is seam length, which involves measuring and verifying the lengths of adjoining seams to ensure they align correctly during sewing. At the garment sample level, size consistency serves as the primary evaluation criterion which is assessed by comparing the sample against size charts while allowing for a permissible margin of error, which varies by garment type and zones. Pattern verification is typically conducted by pattern makers, graders, and quality control (QC) personnel within companies or manufacturing facilities.

(b) Pattern accuracy related to 2D and 3D data

Since the evaluation metrics for pattern accuracy in 2D and 3D body data-based pattern construction share significant commonalities, this section integrates their discussion. The key metrics primarily include edge length (from flattening results) and Panel L2 (from AI-generated patterns) at the pattern level, as well as cross-sectional ease allowance and stress analysis at the garment level.

- Edge length of flattened patterns

For patterns obtained through flattening techniques, accuracy is typically assessed by comparing the edge lengths of 3D garment sections with those of their flattened counterparts. The length change ratio is calculated to identify the most accurate flattened pattern—defined as the one closest to the original 3D edge length on the body surface. This approach has been applied in studies on motorcycle suits[161] and speed skating suits[160], demonstrating its effectiveness in preserving garment geometry during the flattening process.

- Metrics in AI-generated patterns

The accuracy of AI-generated sewing patterns is commonly evaluated through several key metrics.

Panel L2 quantifies shape prediction accuracy by computing the L2 distance between the edge parameters of predicted and ground-truth panels. #Panel and #Edges respectively assess the correctness of the predicted number of garment panels and the accuracy of edge counts within each panel.[170] These metrics collectively ensure the geometric precision and structural reliability of AI-generated patterns.

- Cross-sectional ease allowance

Ease allowances between garments and the body serve as critical indicators for evaluating the fit and comfort of functional clothing, particularly in dynamic postures. The analysis of cross-sectional ease allowance in motion has been a focal point in fit assessment research. For instance, Lee and Park[171] analyzed thigh, knee, and crotch ease variations in firefighter trousers during standing and kneeling postures to identify design improvements. Nam et al.[172] optimized a liquid-cooled vest prototype by scanning wearers in work postures and statistically analyzing the radial distance values and their distribution rates between the body and garment. Similarly, Kim and Baytar[173] compared virtual and actual dynamic fit by examining waist cross-sectional areas.

- Stress analysis

Pressure testing devices and software-based pressure simulations enable the assessment of the relationship between garment stress and pattern design. Bragança et al.[159] evaluated dynamic pattern functionality by placing pressure sensors at high-stress areas on garments, collecting static and dynamic pressure data as subjects performed specified movements. Silina et al.[174] employed a multipoint pressure sensor matrix to measure pressure distribution at specific calf measurement points in both static and dynamic postures, revealing correlations between garment structure (such as pattern design and grading) and pressure distribution. In addition, Teyeme et al.[175] utilized CLO3D pressure maps to examine the influence of cycling postures and fabric performance on pressure distribution. Through multiple pattern adjustments, they ensured that both static and dynamic cycling postures exhibited optimal pressure and strain characteristics, enhancing garment fit and performance.

(c) Pattern accuracy related to 4D data

Utilizing 4D technology to evaluate the ergonomic accuracy of garment patterns has emerged as an innovative approach. Dynamic fit tests provide critical insights into pattern quality by assessing garment performance during movement. These tests involve human subjects performing specific activities, with advanced technologies capturing and analyzing motion-derived data.

- Slippage in motion

Slippage induced by motion and garment stability in dynamic conditions are promising indicators for assessing pattern accuracy. During human movement, garments exhibit a degree of slippage along the body surface, enabling partial adaptation to changes in body dimensions. Displacement at key areas such as cuffs, waistbands, and pant hems under dynamic postures or continuous motion is frequently analyzed. Before the advent of 4D technology, researchers relied on ergonomic experiments to study garment slippage. For instance, Zhang and Little[176] investigated the displacement of key reference points between garments and the human body under dynamic postures. Their findings suggested that smaller displacement indicated better garment-body synchronization, implying that the dynamic ease in the pattern was appropriate. In contrast, larger displacement suggested that the garment was restricted or misaligned with body movement, highlighting potential pattern deficiencies. In addition, virtual try-on has been increasingly utilized to evaluate dynamic fit. Hu et al.[177] established a universal 3D coordinate system to quantitatively assess dynamic air gap thickness and proposed a 3D garment vector field method to measure garment sliding during walking. Moreover, as 4D technology evolves, Klepser et al.[166] applied 4D scanning to examine waistband slippage in male subjects performing a step-on-the-ladder motion. The results demonstrated that excessive slippage led to compromised ergonomic performance, emphasizing the necessity of pattern optimization.

- Point cloud distance or mesh vertex distance

Point cloud distance is an emerging metric for evaluating the relative movement of scanned garments. For example, Muenks et al.[178] employed 4D scan data to assess ergonomic fit under firefighting clothing conditions, utilizing point cloud distances across scan frames to quantify the relative movement of the garments. Zhang et al.[179] compared the differences in mesh vertex distances between 4D scan data of individuals walking in trousers and the walking simulation data in CLO software. This study explored the discrepancies in dynamic effects between real and virtual garments during motion, which may stem from fabric property measurements, limitations in simulation algorithms, and other factors. Overall, slippage and point cloud distance serve as key indicators of pattern accuracy,

reflecting the interaction between the final garment and the body. Effectively utilizing these indicators for pattern modifications and further enhancing pattern accuracy remains a challenging task.

Figure 13 illustrates commonly used pattern accuracy evaluation metrics.

Applications

Fashion product development. Different body data influence how fashion products are conceptualized, designed, and tested. This section focuses on pattern construction applications and garment development methods that have been successfully implemented in commercial practice, without analyzing innovative academic methods that have yet to achieve widespread adoption.

(a) 1D and 2D methods in product development

In the commercial application of 1D body data for pattern-making, parametric pattern design is the most mature and widely adopted method, aligning with the trend of mass customization. Parametric customization and personalized grading are applied extensively in custom menswear, professional attire, and group uniforms, with many custom apparel companies promoting the concept of "one person, one pattern." The application of 2D body data pattern-making technologies primarily involves extracting 1D measurements from 2D body images, which are then used in parametric design and grading. Fundamentally, these commercial practices remain driven by 1D body measurements for personalized pattern generation. The core principle is to utilize tailored measurements of customers, thereby reducing the difficulty of selecting a perfect fit from a limited range of standard sizes, offering products with ideal sizing and design features, and saving significant time and effort otherwise spent on selection and fitting. Numerous brands have adopted this approach. For example, Levi Strauss was one of the earliest brands to introduce mass customization, with its custom-made jeans program offering a personalized fit. Similarly, Brooks Brothers has developed an app-based platform that includes a MTM program for custom tailoring and alterations.

(b) 3D methods in product development

Pattern construction based on 3D body data has become a key focus in the apparel industry, benefiting

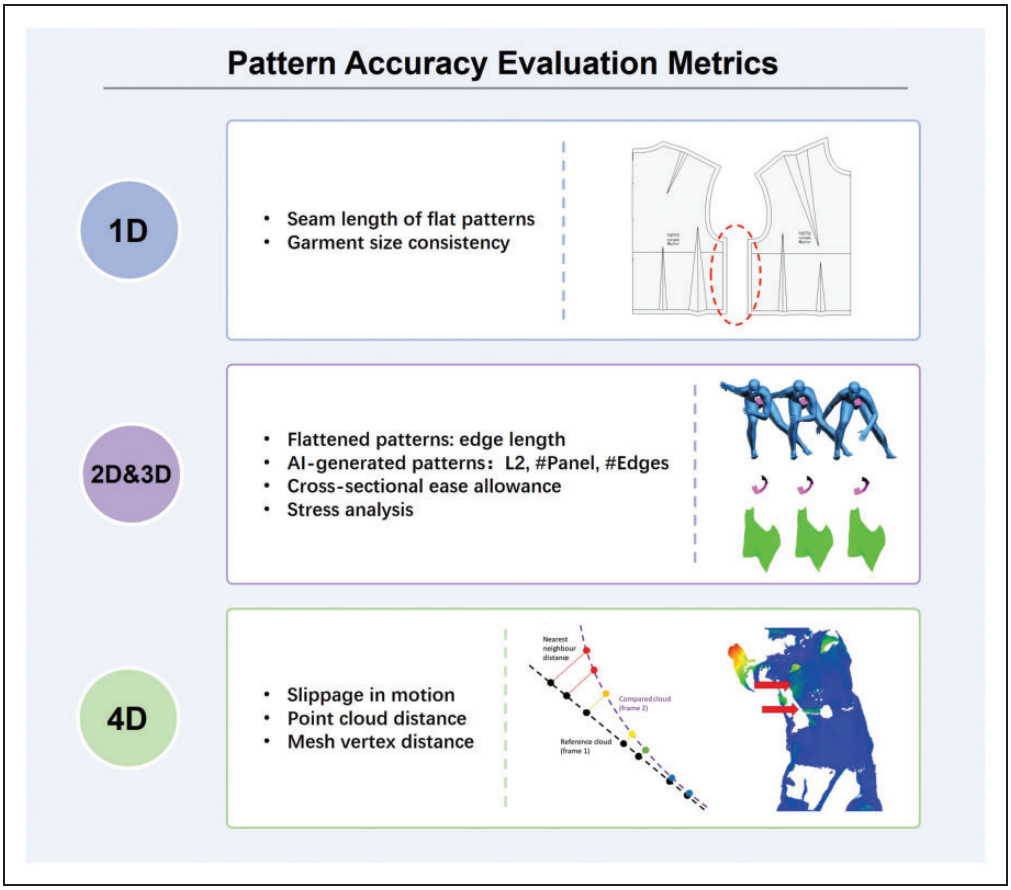


Figure 13. Pattern accuracy evaluation metrics.[160,178]

software developers, fashion companies, independent designers, and students alike. First, 3D body data play a crucial role in the development of virtual fitting software. Leading virtual fitting software providers, such as CLO3D, Style3D, and Browzwear, continuously strive to enhance their physics simulation capabilities while exploring the integration of AI into the garment development process. For instance, CLO3D and Toray's Pattern Magic have developed advanced surface flattening technologies that allow direct extraction of flattened patterns from body surfaces, while CLO3D also offers a user-friendly 3D grading function. In the field of AI-generated patterns, Style3D has experimented with AIG-Pattern using simple dress designs, achieving synchronized editing between 3D garments and 2D patterns. Second, 3D virtual prototyping has been integrated into fashion design education, helping students bridge the gap between 2D pattern making and 3D garment construction. Features such as synchronized 2D patterning and 3D fitting in CAD software, as well as surface flattening functions, assist beginners in understanding the relationship between 2D and 3D. Moreover, 3D virtual design fosters creativity and enhances learning outcomes, equipping students with the digital skills required for the industry's transformation.[180] Lastly, 3D technologies have been widely adopted in fashion companies for both garment evaluation and accelerated workflow. Companies focusing on functionality, such as Nike and Adidas, employ advanced 3D modeling techniques to assess fabric elasticity, compression, and aerodynamic properties before production, ensuring optimal performance while streamlining the design process. Meanwhile, companies prioritizing fast product turnover, such as Zara and H&M, utilize 3D design software to accelerate production cycles, enabling rapid responses to market demand and reducing the waste associated with physical sample production.

(c) 4D methods in product development

4D methods hold significant potential for diverse applications in the apparel industry, particularly in enhancing the design of functional and protective garments for sectors such as sports, military, firefighting, medicine, and space exploration. By optimizing fit, comfort, safety, and performance, it supports the development of compression garments, performance wear, and adaptive clothing.

Primarily, in the sportswear and workwear industries, companies place significant emphasis on the dynamic performance of garments, as the research, development, and testing of sports products are highly rigorous. Leading apparel brands have increasingly integrated 4D data into pattern construction to enhance the design of high-performance garments. First, 4D scanning has been adopted in the industry; for instance, Under Armour has incorporated 3dMD's 4D scanning technology into its human body research and product development,[46] whereas Carhartt, in collaboration with the German research institute Hohenstein, utilizes 4D scanning for dynamic body data collection, garment design, and performance evaluation. The insights derived from these assessments contribute to the optimization of pattern design. Second, motion capture technology plays a crucial role in sportswear development by providing precisely tracked athletic movements for apparel designers, sports coaches, and biomechanics experts. British designer brand Aitor Throup, known for its "anatomical tailoring" approach, employs motion capture technology to analyze muscular stretching and deformation, enabling the selection of suitable elastic fabrics for sportswear. Similarly, Mizuno leverages motion capture analysis to obtain quantitative performance data for sporting goods testing. Lastly, other biomechanical measurement tools, such as plantar pressure devices, further enhance product innovation. For example, Li-Ning, a Chinese sportswear company, has utilized dynamic foot data to design shock-absorbing insoles, and the future integration of plantar pressure measurement with 4D scanning systems could provide a comprehensive analysis of foot movement, improving footwear design and performance significantly.

Online sales and entertainment. Digital garment pattern construction and prototyping are integral to the broader digital transformation, driven by consumers' increasing demand for personalized and immersive shopping experiences. Whether through online pattern customization, virtual try-ons, or the creation of digital fashion assets, these technologies not only deliver captivating digital experiences but also enhance customer engagement, boost sales, and cultivate brand loyalty.

(a) 1D methods in online sales and entertainment

In the commercial sector, 1D body data are primarily utilized for parametric pattern design and sales. Numerous online garment pattern sales platforms worldwide offer parametric basic blocks or programming interfaces, enabling users to design personalized patterns by inputting body measurements and adding design elements such as fit, ease, and dart placement. This approach caters to both beginners and pattern-making experts, as exemplified by platforms such as Sewist CAD and PatternLab.

(b) 2D methods in online sales and entertainment

Since human body cross-sectional data require 3D scanning and specialized software for extraction, they are not easily accessible, limiting the widespread adoption of this approach. In commercial applications, 2D human body image data are predominantly used, with AI models generating 3D garment models for industries such as gaming, film, and e-commerce. For example, the digital fashion platform The Fabricant employs its AI-powered Fabricant Intelligent tool to transform 2D images into immersive 3D assets within seconds, producing high-quality visuals for gaming, VR/AR, interactive shopping, and immersive brand experiences. Similarly, Clothware converts 2D clothing images into realistic 3D garments, enabling character design and seamless integration into e-commerce platforms.

(c) 3D methods in online sales and entertainment

The creation of 3D garment models based on 3D body data and their subsequent commercialization represents one of the most promising applications in the fashion industry today. Although the development and utilization of 3D prototypes primarily commence during the design and manufacturing stages of the product lifecycle, the digital assets generated throughout this process maintain their value even after the product reaches retail shelves and becomes available to consumers.

On the one hand, 3D garment models are employed in digital runway collections and virtual fashion shows, offering innovative approaches to fashion presentation. On the other hand, they serve as 3D digital assets with commercial circulation potential. For instance, numerous online platforms such as The Fabricant, Carlings, and DressX provide opportunities for users to purchase and personalize digital garments for use in AR/VR environments. Some virtual garments are even sold as NFTs (nonfungible tokens), propelling fashion into the digital economy. Furthermore, 3D virtual garments are also utilized for online virtual try-ons. Digital avatars and virtual clothing visualization allow customers to preview garment fit before purchasing, effectively bringing the in-store shopping experience to online platforms.

(d) 4D methods in online sales and entertainment

4D body data holds significant potential in e-commerce applications, particularly in sportswear. Dynamic virtual try-ons allow users to visualize the dynamic effects of wearing garments, thereby reducing high return rates caused by the inability to try on or the limitations of static virtual fitting. For example, Kong et al.[181] found that online customers preferred 4D golf apparel simulations over static images. Further applications could explore the generation of dynamic virtual try-on videos based on user data, providing interactive simulations of garment–body interactions and enhancing the accuracy of online garment customization and purchasing.

Figure 14 illustrates the relevant applications of pattern construction methods.

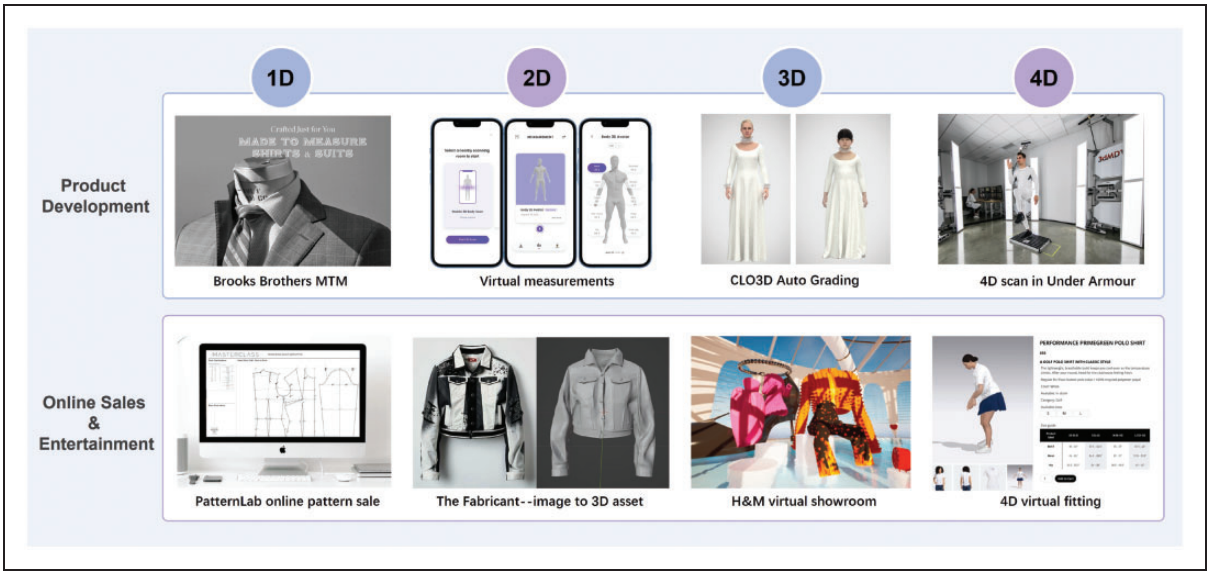


Figure 14. Relevant applications of pattern construction methods.

Ethical considerations

Body data privacy. The collection and utilization of body data in digital fashion pose ethical challenges, particularly regarding privacy, informed consent, and data security. As body data become integral to apparel technology, ensuring robust protection is essential for safeguarding privacy and maintaining consumer trust. Specifically, digital body scanning enables precise measurement for compression garments but raises privacy concerns, particularly with direct scans. An alternative approach involves scanning individuals in loose-fitting clothing and using AI models to infer body shape, balancing accuracy with privacy. Furthermore, to mitigate risks, companies must implement strict data protection measures, ensure transparent data usage, and obtain explicit consumer consent. Advanced encryption and anonymization techniques are critical to preventing breaches and upholding ethical standards in digital fashion.

Ethical AI utilization. The application of AI in garment pattern design raises ethical concerns, particularly regarding dataset bias, creativity limitations, and intellectual property protection. First, AI models often reflect biases in their training datasets, leading to unfair fashion recommendations. Many AI-generated models predominantly feature white female body types, underrepresenting diverse races, body shapes, genders, and ages. To address this, diverse datasets should be used, and ethical guidelines should regulate AI applications in fashion. Moreover, AI lacks human intuition and artistic perception, often producing repetitive or uninspired designs. Over-reliance on AI may limit originality, making it essential for designers to balance AI efficiency with human creativity. In addition, AI-generated garment or pattern designs risk replicating existing works, complicating copyright protection. Without clear legal frameworks, it becomes difficult to define ownership and prevent infringement. Therefore, to ensure ethical AI use, the fashion industry must promote diversity, foster human–AI collaboration, and establish regulatory safeguards.

Prospects and challenges

In-depth exploration of dynamic pattern construction

Future in-depth exploration of dynamic garment pattern design demands higher levels of technological integration and conceptual innovation. Several key points worth noting are listed in the following.

First, beyond the 4D body scanning techniques discussed in this paper, it is essential to integrate 4D data with biomedical and engineering methods. Medical imaging techniques such as magnetic resonance imaging (MRI) and computed tomography (CT) enable the analysis of soft tissue deformation, whereas engineering tools such as 3D-DIC support precise tracking of skin movement and have been applied in the design of mechanical counter-pressure spacesuits. In addition, flexible sensors from intelligent electronics, such as stretchable yarn sensors, provide accurate dynamic body measurements and support 3D reconstruction.[182] Future research should focus on synchronizing dynamic body data with other biometric signals (such as thermal data, force platforms, accelerometer, and electromyography [EMG] devices), for example, examining the correlation between soft tissue vibration and EMG signals during the use of compression garments can offer valuable insights for garment design and evaluation.[183]

Second, advancing the integration of 4D body data with garment design parameters is crucial, such as pattern geometry, fabric properties, manufacturing techniques, and biomechanical factors. For instance, Černiavskaja et al.[184] developed a customized compression leg sleeve with adhesive stripes inspired by Kinesio taping, aimed at enhancing athletic performance by limiting excessive leg flexion, as illustrated in Figure 15(a). The dynamic pattern can be optimized through ergonomic evaluation and simulation modeling to validate its effectiveness.

Third, advancing dynamic modeling of the human body and garments is essential. On a broader scale, national and industrial initiatives should prioritize the development of large-scale dynamic body databases to support research on body–garment interactions. Conversely, at a finer scale, integrating soft tissue mechanics, kinematics, and dynamics into virtual human models enhances simulation realism and movement adaptability. For instance, platforms such as Style3D and Vstitcher have created avatars capable of realistically simulating soft tissue deformation, adjusting body shape in response to garment pressure, localized softness, and gravity, as shown in Figure 15(b).

Developing a dynamic pattern construction methodology is a long-term and challenging task with significant potential for garment applications. The core components, including the acquisition of dynamic body data, the translation of these data into applicable pattern design parameters, and the evaluation of dynamic fit, remain in their early stages. This constitutes a promising and forward-looking direction in the field of apparel technology. Figure 16 presents the theoretical framework for the application of 4D data.

AI-facilitated pattern generation

Generative AI holds significant promise for the future of digital garment pattern design. In response to the

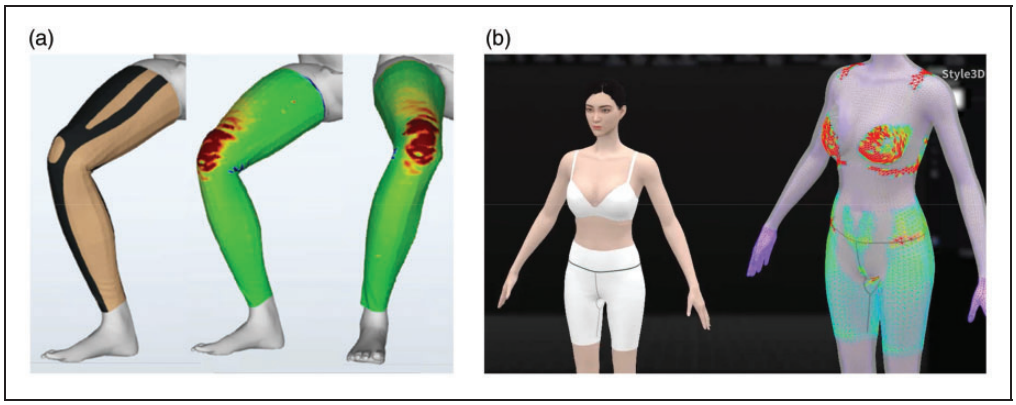


Figure 15. Dynamic design techniques: (a) Kinesio taping theory in compression garment design[184] and (b) Style3D soft mannequin.

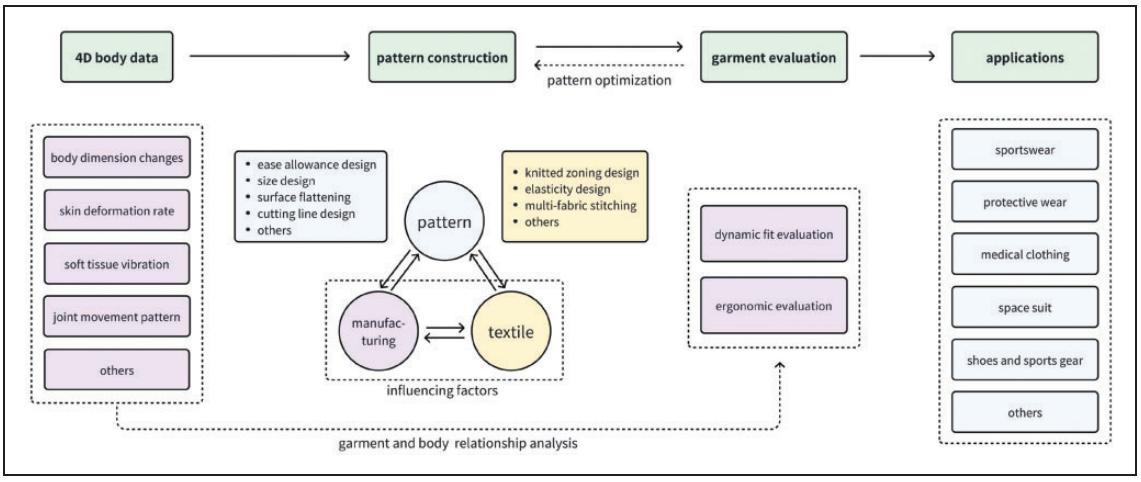


Figure 16. Theoretical framework for the application of 4D data.

three major technical bottlenecks discussed earlier regarding the application of AI in pattern design, this section proposes three corresponding future directions.

First, to address the issue of AI's limited understanding of design language, multimodal learning within DL frameworks presents a promising solution. By incorporating diverse types of input data such as textual descriptions, images, hand-drawn sketches, 3D models, videos, and pattern structures, this approach facilitates the integration of complementary information within the fashion domain. Such integration enhances the model's ability to interpret creative design language from multiple dimensions, thereby improving both the accuracy of garment and pattern generation and the integrity of structural detail representation. This multimodal strategy provides an effective means of overcoming the limitations associated with single-modality learning.

Second, to address the issue of AI's inaccuracy in capturing designers' pattern requirements, AI-aided pattern generation can draw on commercial pattern design practices by incorporating brand style, seasonality, and collection attributes. By distinguishing brand-specific pattern DNA from transient fashion elements, it supports the establishment of a robust parent pattern (block) from which derivative patterns can be systematically developed. This approach helps maintain consistency in pattern design even when the AI model's comprehension is not yet fully developed. Furthermore, enriching the diversity of dataset content and ensuring the accuracy and completeness of annotations are essential. Given the vast variety of garment styles, it is crucial to construct a comprehensive dataset that captures the full spectrum of possible variations. Detailed annotations should encompass style categories, pattern structures, textures, and fabric types, which collectively enhance the model's generalizability and robustness.

Third, in response to the limited practicality of AI-generated patterns and the difficulty of iterative optimization, it is essential to integrate expert knowledge of pattern construction into AI models.

Functional details such as darts, pleats, gathers, wrinkles, and interlaced structures must be represented accurately through specialized algorithms to meet manufacturing requirements. Furthermore, model design should enhance the efficiency of human–AI cocreation. This includes developing intuitive methods for incorporating designers' aesthetic judgments, enabling secondary editing of patterns, and establishing reliable evaluation metrics. Although AI offers powerful computational capabilities, it cannot replace human creativity in garment pattern construction. Future research should therefore focus on integrating data-driven approaches with creative and perceptual thinking to enable more advanced and collaborative design solutions. The application of cognitive computing may further enhance this integration within the design process. Figure 17 presents the theoretical framework for the application of AI tools.

Intelligent integration of CAD systems

The intelligent integration of CAD systems holds significant potential for advancing digital garment pattern design, promoting seamless, efficient, and adaptive workflows.

First, enhancing the convenience of the entire process is crucial. Many CAD systems currently integrate with upstream and downstream tasks related to pattern construction, such as 3D body scanning, pattern design, virtual fitting, grading and marking, cutting and sewing, sales and marketing, and inventory control. For example, systems such as Gerber AccuMark, Optitex, and Lectra are integrated with product lifecycle management systems, enabling companies to remain competitive and responsive to market demands. In addition, research can be further advanced by integrating lifecycle assessment software with CAD technology to facilitate environmental assessment, offering a promising approach for evaluating the environmental impact at the early stages of product design.[185] Coupled with advanced AI technologies, this holistic integration improves both accuracy and efficiency, creating a seamless workflow from initial design to end users.

Second, system and file compatibility are essential for efficient collaboration. The ability to work across different software platforms promotes better communication and smoother workflows. Improving interoperability and seamless data exchange between software minimizes integration issues. Incorporating application programming interface functions across diverse CAD platforms fosters more efficient workflows and optimizes the creation and production processes.

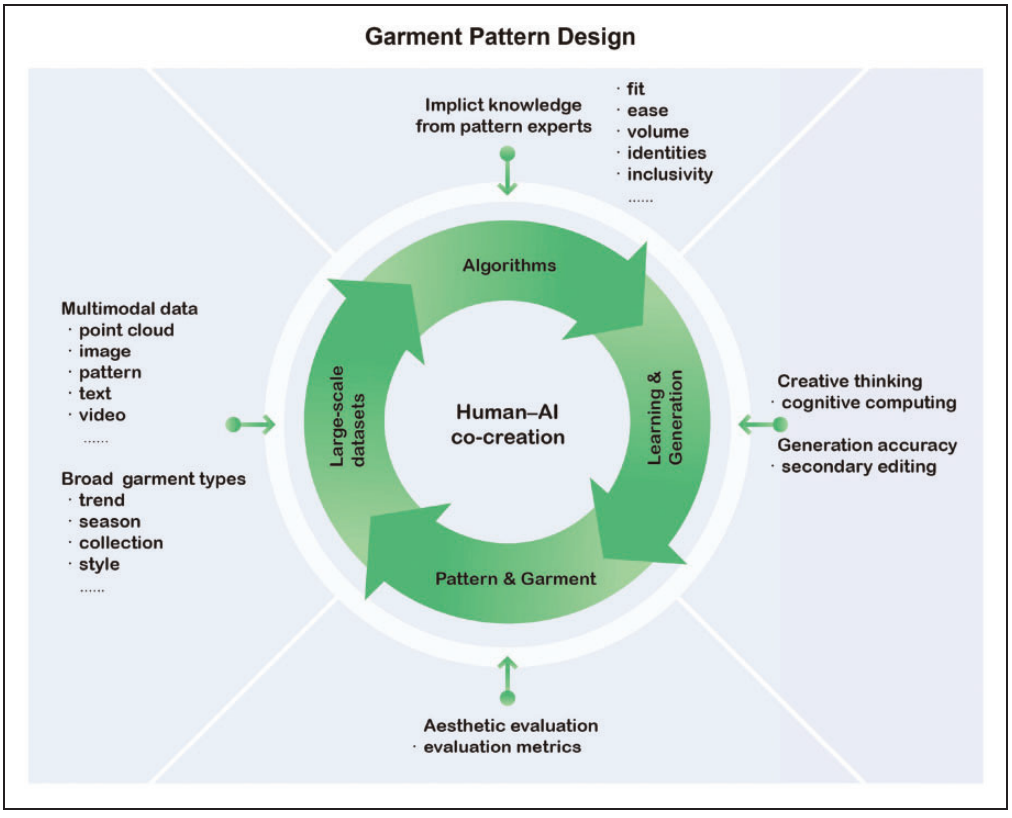


Figure 17. Theoretical framework for the application of AI tools.

Moreover, cloud-based collaborative design is an emerging trend. Cloud computing and remote collaboration capabilities are expected to further enhance efficiency and accessibility in garment pattern construction practices. Such advancements enable seamless collaboration among pattern makers, designers, and other stakeholders, fostering a dynamic and interactive design environment.

Finally, the integration of AR and VR technologies is expected to revolutionize creative practices in digital pattern design. AR/VR-based digital draping enables garment visualization and manipulation without requiring physical fabric or dress forms, significantly reducing time and effort. Compared with conventional 3D CAD systems, AR/VR draping offers enhanced tactile and spatial perception, which is essential for accurate draping practices.

By leveraging these intelligent integration strategies, CAD systems will continue to evolve toward greater dynamism, efficiency, and innovation in digital garment pattern design.

Customization and sustainability

Both the precise anthropometric data and digital garment design tools are crucial in enhancing customization and promoting sustainability within the fashion industry.

4D and AI technologies present unprecedented opportunities for customization, fostering a personalized fit, design, and customer-centric fashion production. From the perspective of 4D technology, advanced body anthropometric techniques, and dynamic data capture provide detailed insights into individual body shapes and sizes, facilitating the creation of garments that offer superior fit across various activities, thereby elevating the level of customization. In terms of AI applications, pattern construction processes are being transformed through the analysis of extensive datasets encompassing sewing patterns, body models, textures, etc. By leveraging sophisticated algorithms, AI can generate bespoke patterns that achieve a more precise fit while accommodating individual aesthetic preferences. In addition, the development of data-sharing and interactive design platforms enables consumers to actively participate in the design process by submitting their measurements and preferences, enhancing user experience and potentially extending the lifespan of garments.

The sustainable significance and trends of digital garment pattern construction are reflected in design, production, sales, and the circular economy. From a design perspective, the adoption of virtual prototyping and iterative design processes significantly reduces the development of physical prototypes, thereby enhancing sustainability by conserving time and resources. In production, manufacturers can leverage data analytics to forecast sizing trends, enabling the production of garments in appropriate quantities and sizes to achieve on-demand manufacturing, which minimizes material waste and prevents overstocking. In the sales sector, the integration of dynamic virtual fitting systems could potentially enhance size accuracy by incorporating pressure mapping, thereby reducing return rates. Lastly, from a circular economy perspective, AI-driven algorithms may facilitate the upcycling and redesign of stock garments by generating feasible transformation solutions, experimenting with sewing patterns, texture, and color combinations from different apparel categories, and exploring the redesign process with various technical and aesthetic solutions.[186] This exploratory approach could empower designers to develop sustainable alternatives and increase garment longevity, fostering responsible practices within the fashion industry.

Conclusion

Garment pattern construction is driven by body data, and the strategic use of these data through digital technologies enhances both the accuracy and efficiency of the process. This review provides a detailed examination of the evolution of body data utilization and the advancement of tools in pattern construction, with a particular emphasis on the future potential of 4D dynamic data in garment pattern studies and applications. Building on this foundation, this review systematically explores pattern construction methods, focusing on how body data across different dimensions can be utilized with advanced tools. In the future, garment pattern construction is expected to increase dynamism, intelligence, integration, and sustainability.

This review aimed to establish a foundation for future research in garment pattern construction by promoting the exploration of dynamic body data and the application of AI. As 4D data become more widely adopted in the fashion industry, the development of dynamic anthropometric databases is becoming increasingly achievable. These databases will support highly customized fitting and improved ergonomic performance in garment design, with implications for fields such as ergonomics, kinesiology, healthcare, and aerospace. AI, informed by data-driven insights, will enhance decision-making and enable the generation of personalized pattern designs. The integration of AI and 4D technology offers significant potential to optimize production processes and advance performance-oriented apparel development in the digital era.

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